

Electoral Equilibrium

Task Instructions & Hour Ledger

320 Hours · 8 Weeks · 40 hrs/week

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The Big Idea

Electoral forecasting is static. Polling aggregators tell you where the race stands today but cannot answer: *“If **X** happens tomorrow, which voter groups must shift for the Democrat to still win?”*

Electoral Equilibrium answers in real time. Select a party (Democrat or Republican), type any hypothetical event — a scandal, an assassination, an October Surprise — and the app immediately shows how that party’s coalition of ten key voter groups (African Americans, Latinos, Asians, Seculars, Jews, Muslims, Women, Catholics, Protestants, Evangelicals) must rebalance, and the probability of achieving it.

Built with a fine-tuned Mistral 7B LLM, a mean-variance stochastic optimizer, and a two-stage NLP pipeline drawing on social media bio inference — grounded in two decades of Prof. Espinosa’s research on religion, race, and presidential voting.

How to Use This Document

Each task entry shows: a concrete deliverable, the files it touches, and its hour budget. Hours are set conservatively — if a task runs short, bank the time and use it as flex on the next heavy day. Every week closes with a **pull request** to main and a **brief weekly supervision check-in** with Prof. Espinosa.

Symbol	Meaning	Note
[N hrs]	Hour budget for this task	Running total per day = 8 hrs
☐	Checkbox to mark complete	Print and tick off
HPC	Run on USC Laguna cluster (compute); CMC HPC for hosting only	Submit batch job, don’t block locally
M5	Run locally on MacBook Pro	Fast iteration / experimentation
PR	Pull request to main	Required before week closes

Hour Summary by Week

Week	Focus	Hours
1	Infrastructure & Artifact Contracts	40
2	Historical Data Pipeline	40
3	Baseline Voter Portfolio	40
4	RoBERTa Pipeline & Fine-Tune Dataset	40
5	LLM Fine-Tuning & Inference Endpoint	40
6	Stochastic Optimization & Monte Carlo	40
7	Web Application	40
8	Hardening, Validation & Deliverables	40
Total		320

AI Research Assistant Setup (Pre-Week — 4 hrs)

Do this before the first day of work. The goal is a 24/7 AI research assistant running on the Raspberry Pi 5 that you can message from Telegram or WhatsApp at any time. Routine queries (paper summaries, quick code questions, daily briefings) route to **Gemini Pro** for free. Deep architectural work routes to **Claude** on demand.

Component	Role	Cost
OpenClaw on Raspberry Pi 5	Gateway, memory, scheduling	Free (open-source)
Google Gemini Pro (AI Studio API)	Lightweight daily queries	Free tier / Pro sub
Anthropic Claude API	Deep reasoning and code	Pay per token
Telegram or WhatsApp	Chat interface	Free

Day 0 (Sun — Environment Setup: OpenClaw, Syncthing & Pi NPU (pre-work, 6 hrs))

□ **Task:** Apply for student programs and free tiers **before anything else**. (i) **GitHub Student Developer Pack** at education.github.com/pack — unlocks JetBrains IDEs, Namecheap domain (.me free for 1 year), Canva Pro, and more. Takes minutes. (ii) **Social media access — entirely free, no subscription needed.** Sign up for **Bluesky** at bsky.app (primary live collection, AT Protocol firehose, no cap) and **Apify** at apify.com (secondary live collection, free tier 500 results/run, covers the X demographic profile without any API cost). The bulk of historical Twitter data comes from free public archives downloaded in Week 4: 56M Kavanaugh tweets, 2020 election datasets, MeToo tweets, and more. **Do not sign up for the X Basic API (\$100/month) — the archives are superior for historical shocks and Bluesky/Apify covers live collection at zero cost.** (iii) **Meta Content Library** at developers.facebook.com/docs/content-library — free academic access; takes 2–4 weeks. Apply today. (iv) **Modal account** at modal.com — \$30/month free credit; at Mistral 7B on an A100 (approximately \$0.0003 per call), covers around 100,000 inference calls for free. Sign up immediately. [1 hrs]

□ **Task:** SSH into the Raspberry Pi 5. Install OpenClaw: `curl -fsSL https://openclaw.ai/install.sh | bash`. Run `openclaw onboard` and follow the wizard. Select Telegram or WhatsApp as the primary channel. Configure Gemini as default model and Claude Sonnet as the `@claude` route. Store both API keys in `/etc/environment`. Docker-containerise OpenClaw for security. [1 hrs]

□ **Task:** Install **Syncthing** on the four local machines only (M5, Intel Mac, Pi). **Do not install Syncthing on USC Laguna or CMC HPC** — HPC nodes kill background daemons and a Syncthing daemon will be terminated silently, starving any SLURM jobs that depend on it. Create a shared `electoral-sync/` folder synced across the four local devices. Subdirectories: `rawdata/` (collected posts and articles), `artifacts/` (pipeline outputs), `data/finetune/` (training data). Test by writing a file on the Intel Mac and confirming it appears on the M5 within 30 seconds. HPC data transfer uses explicit `rsync/scp` before each job submission. [1 hrs]

□ **Task: Compile the Hailo HEF now in Week 0 — not Week 4.** NPU compilation to custom silicon routinely causes driver mismatches, ONNX graph translation failures, and quantization errors. This is a multi-day risk if hit in Week 4. Eliminate it now. (i) Install the Hailo Dataflow Compiler and SDK: `pip install hailo-sdk-client hailo-model-zoo onnx sentence-transformers`. Verify: `hailortcli fw-control identify` should return device info. (ii) Export all-MiniLM-L6-v2 to ONNX. (iii) Run `hailo optimize minilm.onnx --hw-arch hailo8l`. (iv) Run `hailo compile --hw-arch hailo8l -o models/minilm.hef`. (v) Verify: `from hailo_platform import HEF; HEF("models/minilm.hef")`. **CPU fallback (mandatory):** if compilation fails for any reason, set `"pi_npu_enabled": false` in `configs/base.json`. The bio classifier runs on Pi CPU at ~20ms/bio instead of sub-ms — slower but correct. The rest of the pipeline is unchanged. Week 4 must not be blocked on NPU compilation under any circumstances. [3 hrs]

□ **Task:** Set up the **Intel Mac** as the social media collection machine. Configure both free collectors:

- **Bluesky:** install `atproto` (`pip install atproto`). Test `app.bsky.feed.searchPosts` with the Ayatollah keywords. Confirm author `description` and `langs` fields are returned.
- **Apify:** install `apify-client` (`pip install apify-client`). Set the API token in `.env`. Test the X Scraper actor with the Ayatollah keywords. Confirm output matches the same `posts.jsonl` schema as `BlueSkyCollector`.

Set up a **launchd plist** so both collectors restart on reboot. `BlueSkyCollector` runs continuously (no cap). `ApifyCollector` is triggered per-shock (500 results/run free tier). Both write to `rawdata/social/{platform}/{shock_id}/posts.jsonl`. **Do not set up the X Basic API — it is not used in this project.** [2 hrs]

□ **Task:** Set up the **Intel Mac** for news scraping (previously Windows role): Python 3.11, `jupyter`, `pandas`, `matplotlib`, install Python 3.11, project dependencies, and `Synthing`. Configure `Synthing` to sync the shared folder. Confirm Jupyter can read files from `electoral-sync/rawdata/`. [1 hrs]

Parallel machine strategy for Weeks 4–5: Intel Mac runs Bluesky collector continuously and Apify per-shock. Pi NPU classifies bios in real time as posts arrive via `Synthing`. Intel Mac runs the news scraper. HPC scores all collected text via SLURM array jobs. M5 does interactive development. No machine is idle.

Routing strategy for the 8 weeks: Use Gemini for paper summaries, quick questions, drafts, and anything you would normally Google. Switch to `@claude` for pipeline architecture, LaTeX writing, debugging complex logic, and anything requiring the full project context. This keeps your Claude Max usage reserved for real development sessions.

1 Week 0 — Infrastructure & Artifact Contracts (40 hrs)

Goal. By Friday, the pipeline skeleton exists, every stage has a typed artifact contract, and round-trip serialization tests pass. No economic logic is implemented this week.

End condition.

- ☐ `pytest` passes with zero failures.
- ☐ All payload classes serialize and deserialize without data loss.
- ☐ Smoke pipeline runs and writes one artifact file per stage.
- ☐ PR merged to `main`.

Day 1 (Mon — Repo Setup & Project Scaffolding)

☐ **Task:** Create the GitHub repository `electoral-equilibrium`. Initialize with `README.md`, `.gitignore` (Python), and `pyproject.toml`. Add project description, license (MIT), and a brief abstract in the README. [1 hrs]

☐ **Task:** Apply for Meta Content Library access now at `developers.facebook.com/docs/content-library`. Approval takes 2–4 weeks and is required for Week 4 Facebook data collection. Fill in the academic research use-case form: describe the electoral shock sentiment analysis project, cite Prof. Espinosa as faculty mentor, and list CMC as the institution. [1 hrs]

☐ **Task:** Set up the Python environment. Create a virtual environment, install core dependencies: `numpy`, `pandas`, `scipy`, `scikit-learn`, `cvxpy`, `pytest`, `black`, `ruff`. Pin all versions in `pyproject.toml`. [1 hrs]

☐ **Task:** Create the full directory skeleton from the appendix of the development plan. Every folder and empty `__init__.py` file should exist. Commit as `chore: scaffold repository structure`. [1 hrs]

☐ **Task:** Write `AGENTS.md`: contributor conventions, branch naming rules (`feature/weekN-description`), PR format, and the pull-request checklist from the development plan. [1 hrs]

☐ **Task:** Write `DOMAIN.md`: the specification for instantiating this pipeline in a new domain. This is the generalization contract — the document that makes the codebase a platform rather than a one-off tool. Required sections: (i) *What is a domain* — define stakeholder groups, loyalty metric, win condition, and equilibrium threshold; (ii) *Config fields to set* — list every field in `base.json` that is domain-specific; (iii) *Data to provide* — historical event-response dataset format, prediction market contract mapping file; (iv) *Fine-tuning schema* — confirm the JSONL format is domain-agnostic; (v) *Example: electoral vertical* — the current implementation as a worked example. The electoral build is instance 1 of a

general framework; `DOMAIN.md` makes that explicit from day one. [1 hrs]

□ **Task:** Create `electoral/core/types.py` with all type aliases: `Cycle`, `BlocId`, `Weight`, `ElasticityScore`, `ShockDelta`, `LoraRank`. Write a short docstring for each explaining the convention (format, valid range, rationale). [1 hrs]

□ **Task:** Create `configs/smoke.json` and `configs/base.json`. Smoke config uses two blocs, two cycles, and a tiny dataset path so the full pipeline can run end-to-end in under 10 seconds for testing. [1 hrs]

□ **Task:** Create `electoral/core/rng.py`: implement `derive_seed(tokens)` that hashes a list of string tokens into a reproducible integer seed, and `make_rng(seed)` that returns a seeded `np.random.Generator`. Write three unit tests in `tests/test_rng.py` confirming (i) identical inputs produce identical seeds, (ii) order matters, (iii) the generator produces identical draws across calls with the same seed. [1 hrs]

Day 2 (Tue — Schema Helpers & Artifact Envelope)

□ **Task:** Implement `electoral/core/schema.py` with all five validation helpers: `assert_required_keys`, `assert_unique`, `assert_sorted_unique`, `assert_valid_share`, `assert_shares_sum_to_one`. Every helper must raise `ValueError` with a message naming the class and field. No external dependencies. [3 hrs]

□ **Task:** Write `tests/test_schema.py`. For each helper, write at least two positive tests (valid inputs pass silently) and two negative tests (invalid inputs raise `ValueError` with an informative message). Run `pytest -v` and confirm all pass. [2 hrs]

□ **Task:** Create `electoral/core/io.py` implementing the Parquet + JSON artifact I/O layer (Option A). For the envelope (`stage`, `run_key`, `metadata`): `write_json(path, payload)` and `read_json(path)` using `indent=2`, `sort_keys=True`. For tabular data payloads (voter panel, scored posts, fine-tuning dataset): `write_parquet(path, df)` using `engine="pyarrow"`, `compression="snappy"` and `read_parquet(path)` returning a `pd.DataFrame`. Non-tabular payloads (optimizer output, simulation results) remain JSON. `write_artifact(path, envelope, df=None)` dispatches automatically. Write four tests: (i) JSON round-trip, (ii) Parquet round-trip with a 100-row fixture `DataFrame`, (iii) Parquet output is smaller than equivalent JSON (`assert parquet_size < json_size`), (iv) reading a non-existent path raises `FileNotFoundError`. [3 hrs]

□ **Task:** Define the artifact envelope in `electoral/artifacts.py`. Every artifact must contain `stage`, `run_key`, `metadata`, and `data` fields. Write `StageArtifact` as a frozen `dataclass` with `to_dict` and `from_dict`. [1 hrs]

Day 3 (Wed — Payload Classes (Part 1))

□ **Task:** Implement `VoterPanelData` in `electoral/artifacts.py`. Fields: `cycles`, `blocs`, `columns`, `n_rows`, `source`. Implement `to_dict`, `from_dict`, `validate` using schema helpers. Validation must check: `cycles` sorted unique, `blocs` unique, `n_rows` non-negative. [2 hrs]

□ **Task:** Implement `BaselinePortfolioData`. Fields: `method`, `party`, `weights`, `mu`, `target`. `party` is "democrat" or "republican" and must be validated as such. `weights[bloc_id]` is the optimized coalition weight in $[0, 1]$; must sum to 1.0. `mu[bloc_id]` is the historical mean within-bloc vote share *for the selected party* in $[0, 1]$. Validation: `weights` sum to 1.0 within `tol=1e-6`, all weight and `mu` values in $[0, 1]$, both dicts have identical key sets matching the 10 canonical bloc IDs, `party` is one of the two allowed values. [2 hrs]

□ **Task:** Implement `SentimentData`. Fields: `model`, `shocks`, `scores`. Validation must check: `shocks` unique, all score values in $[-1, 1]$, every bloc key in `scores` maps to a dict keyed by all shocks. [2 hrs]

□ **Task:** Implement `LLMFineTuneData`. Fields: `base_model`, `lora_rank`, `n_examples`, `cycles_used`, `adapter_path`. Validation: `lora_rank` positive, `n_examples` positive, `cycles_used` sorted unique. [2 hrs]

Day 4 (Thu — Payload Classes (Part 2) & Round-Trip Tests)

□ **Task:** Implement `ShockResponseData`. Fields: `shock`, `cycle`, `deltas`, `covariance`, `source`. `deltas[bloc_id]` is the LLM-estimated change in within-bloc Democratic vote share ($\Delta\mu_i$). `source` must be one of "llm_unified", "roberta_news_only", "roberta_social_only" — required for the three-way baseline comparison. Validation: all delta values finite and in $[-0.15, 0.15]$, `covariance` is square with side matching number of bloc keys, `source` non-empty string. [2 hrs]

□ **Task:** Implement `EquilibriumData`. Fields: `method`, `party`, `shock`, `weights`, `mu_shifted`, `feasible`, `target_met`, `target`. `weights[bloc_id]` is the rebalanced coalition weight $\tilde{w}_i$; `mu_shifted[bloc_id]` is the post-shock within-bloc vote share $\tilde{\mu}_i^{(P)}$ for the selected party. Both are required by the frontend to display coalition weight shifts and vote-share loyalty shifts for all 10 blocs. Validation: `weights` sum to 1.0, all weight and `mu_shifted` values in $[0, 1]$, `party` is valid, both dicts have identical key sets. [2 hrs]

□ **Task:** Implement `SimulationData`. Fields: `n_simulations`, `seed`, `win_probability`, `percentiles`. Validation: `win_probability` in $[0, 1]$, all percentile lists have exactly 5 values (`p5`, `p25`, `p50`, `p75`, `p95`) each in $[0, 1]$. [1 hrs]

□ **Task:** Implement `MetricsTablesData`. Fields: `tables`. Validation: `tables` keys non-empty strings, values JSON-serializable (call `json.dumps` inside `validate`). [1 hrs]

□ **Task:** Write `tests/test_artifact_roundtrip.py`. Implement the `assert_roundtrip(obj)` helper. Write one round-trip test per payload class using the minimal instances from the development plan. Add at least two negative tests: duplicate cycles in `VoterPanelData`, weights not summing to 1 in `BaselinePortfolioData`. [3 hrs]

Day 5 (Fri — Pipeline Skeleton, Smoke Run & PR)

□ **Task:** Implement `electoral/config.py`: `PipelineConfig` dataclass that loads `smoke.json` or `base.json`. Required fields: `run_key`, `seed`, `data_path`, `blocs`, `party` ("democrat" or "republican"), `target` (equilibrium threshold: Democrat ≈ 0.535 , Republican ≈ 0.520), `output_dir`. Validate that `party` is one of the two allowed values and that `target` is in (0.5, 0.7). Add "party": "democrat", "target": 0.535 to `smoke.json` and `base.json`. Implement `derive_seed` using `rng.py`. [2 hrs]

□ **Task:** Create `electoral/stages.py`: one stub function per stage (`build_voter_panel`, `build_baseline_portfolio`, `build_sentiment_data`, `build_llm_finetune`, `build_shock_response`, `build_optimization`, `run_simulations`). Each stub accepts `PipelineConfig`, constructs a minimal placeholder payload using the correct dataclass, wraps it in a `StageArtifact`, and writes it to disk via `io.write_artifact` (not `write_json` — tabular payloads go to Parquet). Every week replaces its stub with real logic. [2 hrs]

□ **Task:** Write a `Justfile` at the repo root (install just via `brew install just`). Required targets: `smoke` (runs pipeline on `configs/smoke.json`), `test` (runs `pytest tests/ -v`), `score` (submits RoBERTa SLURM array), `train` (submits HPC fine-tuning job), `deploy` (runs modal deploy `deploy/modal_app.py`), `sample` (submits `scripts/sample_archives.py` as a SLURM array to draw stratified samples from raw archives), `clean` (runs `scripts/clean_with_llm.py` locally on the sampled dataset using Gemini Pro), and `continuous` (runs `prefect run pipeline.py --config configs/continuous.json` for nightly incremental updates post-SRP). This saves repeated long commands and eliminates flag typos across 8 weeks of daily use. [1 hrs]

□ **Task:** Implement `electoral/core/io.py`: `write_artifact(path, envelope)` writes the JSON envelope; for tabular data payloads (panel, scored posts, fine-tuning dataset) writes the data as Parquet via `pandas.DataFrame.to_parquet()`. `read_artifact(path)` reads the envelope and auto-detects Parquet vs JSON for the data payload. Parquet is 5–10× faster than JSON for large tabular data and smaller on disk (Syncthing benefit). Confirm round-trip: write a 1,000-row toy `DataFrame`, read it back, assert equality. [2 hrs]

□ **Task:** Replace `electoral/pipeline.py` with a **Prefect flow**. Install `prefect`. Convert each stage function into a `@prefect.task` with `retries=2` and `retry_delay_seconds=30`. Wrap the full pipeline in a `@prefect.flow`. Benefits: automatic retry on failure, parallel task execution where tasks are independent, a visual dashboard (`prefect server start`) showing

stage completion status, and persistent state so a crash restarts from the last successful stage. Run `prefect server start` locally and confirm the smoke pipeline appears in the dashboard. [2 hrs]

☐ **Task:** Weekly PR: open `feature/week1-artifact-contracts` against `main`. PR description must include: summary of what was implemented, the pre-submission checklist from `AGENTS.md`, and any design choices made. Run all tests locally and merge after they pass. [1 hrs]

☐ **Task:** Brief supervision check-in with Prof. Espinosa: confirm the canonical bloc names and equilibrium threshold $V_{eq} = 0.535$. This is a domain confirmation only — Juhi drives the agenda. Document decisions in `DECISIONS.md`. [1 hrs]

2 Week 1 — Historical Data Pipeline (40 hrs)

Goal. By Friday, the data kernel ingests raw survey exports from all five sources and produces a validated `VoterPanelData` artifact.

End condition.

- ☐ Panel passes all invariants: unique tuples, sorted, finite values in $[0, 1]$.
- ☐ Toy fixture (`toy_panel.csv`) runs through the full cleaning pipeline.
- ☐ All data tests pass.
- ☐ PR merged.

Day 1 (Mon — Data Source Audit & Fixture Creation)

☐ **Task:** Download and inspect raw data from all five sources: ARDA, GSS, Gallup, NEP, and Pew. For each source, document: available years, column names, demographic breakdowns, and any known data quality issues. Write findings in `rawdata/SOURCE_NOTES.md`. [3 hrs]

☐ **Task:** Create `tests/fixtures/toy_panel.csv`: a minimal 20-row CSV with columns `cycle`, `bloc`, `vote_share`, `turnout`, `source`. Cover all 10 blocs across two cycles (2016, 2020). Include one intentional dirty row (missing value, wrong type) to test cleaning logic. [2 hrs]

☐ **Task:** Identify and document the column name mapping for each real data source to the canonical panel schema. Store as `rawdata/column_maps.json`: a dict mapping source name to a dict of `{raw_col: canonical_col}`. [3 hrs]

Day 2 (Tue — Loaders & Panel Utilities)

☐ **Task:** Implement `electoral/data/loaders.py`: `load_csv_panel(path)` that reads a CSV with explicit dtypes (`cycle` as int, `bloc` as str, `vote_share` and `turnout` as float64). No date inference. No random sampling. Returns a `pd.DataFrame`. [2 hrs]

☐ **Task:** Implement `electoral/data/panel.py`: `validate_panel(df, required_cols, context)` that enforces all five invariants from the development plan. Each violated invariant must raise a `ValueError` naming the class and field. Write five unit tests — one per invariant. [3 hrs]

☐ **Task:** Implement source-specific loader wrappers in `electoral/data/loaders.py`: `load_arda`, `load_gss`, `load_gallup`, `load_nep`, `load_pew`. Each applies the correct column map from `column_maps.json` and returns a raw (uncleaned) `DataFrame`. [3 hrs]

Day 3 (Wed — Cleaning Logic)

□ **Task:** Implement `electoral/data/cleaning.py: clean_raw_panel(df)` that applies all required cleaning steps: coerce `cycle` to int YYYY, `bloc` to lowercase snake-case str, `vote_share` and `turnout` to float64, drop rows with missing required values (log how many dropped), sort by (`cycle`, `bloc`), raise on duplicate (`cycle`, `bloc`, `source`) tuples rather than silently aggregating. [3 hrs]

□ **Task:** Implement bloc name normalization in `cleaning.py: normalize_bloc(raw_name)` that maps source-specific demographic labels to the canonical bloc identifiers confirmed by Prof. Espinosa (e.g., "White Evangelical" → "evangelical"). Build the map from `DECISIONS.md`. Raise on unrecognized bloc names. [3 hrs]

□ **Task:** Write `tests/test_data_cleaning.py`. Test: (i) clean path with toy fixture produces correct dtypes and sorting, (ii) dirty row is dropped and logged, (iii) unrecognized bloc name raises informatively, (iv) duplicate tuple raises rather than silently aggregating. [2 hrs]

Day 4 (Thu — Data Kernel & Multi-Source Merge)

□ **Task:** Implement `electoral/kernels/data.py: build_voter_panel(config)` that loads all five sources, applies source-specific loaders, concatenates, cleans via `clean_raw_panel`, validates via `validate_panel`, constructs a `VoterPanelData` payload, and returns both the payload and the cleaned DataFrame. Wire into `stages.py` replacing the Week 1 stub. [3 hrs]

□ **Task:** Implement multi-source conflict resolution: when the same (`cycle`, `bloc`) pair appears in multiple sources with different `vote_share` values, compute the weighted average (weight = inverse standard error if available, else equal weight). Document the rule in `DECISIONS.md`. [3 hrs]

□ **Task:** Run the full data kernel on real data for all cycles 2000–2024. Inspect the output DataFrame. Log counts per source per cycle. Flag any bloc-cycle pairs with missing data across all sources. Document gaps in `rawdata/SOURCE_NOTES.md`. [2 hrs]

Day 5 (Fri — Integration, Visualization & PR)

□ **Task:** Write a data audit script `scripts/audit_panel.py` that loads the cleaned panel and prints: (i) coverage matrix (bloc × cycle showing which cells have data), (ii) vote-share summary statistics per bloc, (iii) any obvious outliers (vote share < 0.05 or > 0.95). Save output as `rawdata/audit_report.txt`. [2 hrs]

□ **Task:** Manually cross-check 10 spot values from the panel against the original source documents. Record the spot-check results in `rawdata/spot_check.md`. Flag any large discrepancies for the supervision check-in. [2 hrs]

□ **Task:** Weekly PR: `feature/week2-data-pipeline`. Ensure all tests pass, the audit report is committed, and the PR description summarizes any data quality decisions made. Merge after tests pass. [1 hrs]

□ **Task:** Brief supervision check-in with Prof. Espinosa: share the coverage matrix and get domain input on any missing bloc-cycle cells. Juhi decides and implements the imputation strategy; document in `DECISIONS.md`. [2 hrs]

□ **Task:** Implement the agreed imputation strategy (likely: linear interpolation for consecutive missing cycles, or carry-forward for isolated gaps) in `cleaning.py`. Add unit tests. [1 hrs]

3 Week 2 — Baseline Voter Portfolio (40 hrs)

Goal. By Friday, the ML pipeline produces a validated `BaselinePortfolioData` artifact representing the optimal steady-state voter equilibrium free of shocks.

End condition.

- ☐ Baseline weights sum to 1.0 and every weight is in $[0, 1]$.
- ☐ The portfolio achieves $\mathbf{w}^\top \boldsymbol{\mu}^{(P)} \geq V_{\text{eq}}^{(P)}$ on historical winning cycles for the configured party.
- ☐ Leave-one-cycle-out validation documented for both parties.
- ☐ PR merged.

Day 1 (Mon — Moment Estimation)

☐ **Task:** Implement `electoral/models/ml.baseline.py: estimate_moments(df, party)` that computes $\boldsymbol{\mu}^{(P)}$ (mean within-bloc vote share for party P across winning cycles) and $\boldsymbol{\Sigma}$ (empirical covariance across cycles). For `party="democrat"` use the `vote_share` column directly; for `party="republican"` use `1 - vote_share`. Define “winning cycle” as cycles where the selected party’s candidate exceeded 50% of the two-party vote share nationally — this is party-configurable. Apply a PSD safeguard to $\boldsymbol{\Sigma}$: add ϵI if the minimum eigenvalue is negative. [3 hrs]

☐ **Task:** Write helper `psd_repair(cov, eps=1e-6)` in `ml.baseline.py` that adds ϵI to a matrix until it is positive semi-definite. Write a unit test confirming the output is PSD for a known near-singular matrix. [2 hrs]

☐ **Task:** Log and inspect the estimated $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. Print bloc names alongside their mean vote shares. Cross-reference against Prof. Espinosa’s published research (e.g., Evangelicals near 24%). Flag any large discrepancies for the supervision check-in. [3 hrs]

Day 2 (Tue — CVXPY Baseline Optimization)

☐ **Task:** Implement `electoral/portfolios/cvx.py: solve_baseline(mu, cov, target)` that formulates and solves the constrained quadratic program: $\min_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$ subject to $\sum w_i = 1$, $w_i \geq 0$, $\mathbf{w}^\top \boldsymbol{\mu} \geq V_{\text{eq}}$. Use `cvxpy.Problem`. Return weights as a dict keyed by bloc name. Raise informatively if problem is infeasible. [3 hrs]

☐ **Task:** Implement solver fallback in `cvx.py`: if CVXPY returns infeasible, relax the equilibrium constraint by 1% increments and retry up to 5 times. Log each relaxation. If still infeasible after 5 retries, raise `ValueError` with the final constraint value attempted. [2 hrs]

☐ **Task:** Write `electoral/portfolios/weights.py: equal_weight_baseline(blocs)` and

`value_weight_baseline(df, blocs)` as simple benchmark alternatives to the CVXPY optimizer. These serve as sanity checks and paper baselines. [3 hrs]

Day 3 (Wed — Gaussian Process Classifier)

□ **Task:** Implement the **Gaussian Process** win-probability classifier in `ml_baseline.py` (replacing XGBoost; Option E). Use `sklearn.gaussian_process.GaussianProcessClassifier` with a composite kernel: `RBf(length_scale=1.0) + Matern(length_scale=1.0, nu=1.5)`. Feature matrix X : per-bloc vote shares, turnout rates, cycle index (numeric, so the GP sees temporal ordering). Target y : binary win. The GP provides calibrated posterior predictive probabilities and explicitly models temporal correlations — the 2020 cycle is weighted more heavily than 2000 when predicting 2024. Extract the posterior standard deviation per held-out cycle and store in `artifacts/baseline_loco.json` alongside accuracy. These uncertainty estimates feed the paper’s epistemic humility framing. [4 hrs]

□ **Task:** Write `tests/test_baseline.py`: (i) `estimate_moments` returns correct shapes, (ii) `psd_repair` output is PSD, (iii) `solve_baseline` weights sum to 1.0, (iv) GP classifier outputs calibrated probabilities in $[0, 1]$, (v) infeasible QP raises correctly. [4 hrs]

Day 4 (Thu — Kernel Integration & Validation)

□ **Task:** Implement `electoral/kernels/baseline.py`: `build_baseline_portfolio(config, panel_df)` that orchestrates moment estimation, CVXPY optimization, and payload construction. Wire into `stages.py` replacing the Week 1 stub. [3 hrs]

□ **Task:** Run leave-one-cycle-out (LOCO) validation: for each cycle c , train on all other cycles, predict win probability on c , compare to ground truth. Record accuracy in `artifacts/baseline_loco.json`. **Also** compare the GP’s predicted win probability against the IEM/PredictIt market-implied price at the time of that election (for cycles 2004+). If the model diverges from the market by more than 10 percentage points on a held-out cycle, investigate the moment estimates and constraint bounds before proceeding to Week 4 — a large divergence signals a calibration problem, not just model uncertainty. [4 hrs]

□ **Task:** Inspect the optimized weights. Cross-reference high-weight blocs against Prof. Espinosa’s published demographic research. Write a 1-page interpretation in `docs/baseline_interpretation.md`. [1 hrs]

Day 5 (Fri — Constraints, Edge Cases & PR)

□ **Task:** Implement `electoral/portfolios/constraints.py`: `ConstraintSpec` dataclass specifying simplex constraints plus any user-defined per-bloc bounds (e.g., Evangelicals must be at least 20%). Integrate into `solve_baseline`. [2 hrs]

□ **Task:** Add edge case handling: (i) single-bloc input, (ii) all blocs equal weight produces feasible solution, (iii) target > 1.0 raises immediately without calling solver. Add tests for each. [2 hrs]

□ **Task:** Weekly PR: `feature/week3-baseline-portfolio`. Include the LOCO validation results and interpretation doc in the PR. Merge after tests pass. [1 hrs]

□ **Task:** Brief supervision check-in with Prof. Espinosa: share the baseline weights and LOCO results. Get domain feedback on whether the demographic ordering aligns with his published research. Juhi decides on any constraint adjustments and documents in `DECISIONS.md`. [3 hrs]

□ **Task:** Implement `electoral/kernels/raking.py`: marginal calibration (iterative proportional fitting) as an optional post-processing step on the stratum weights from `build_constraint_spec`. The algorithm: (1) rescale race weights to sum to 1 matching voter panel race marginals; (2) rescale religion weights matching religion marginals; (3) rescale gender weights matching gender marginals; repeat until convergence (tolerance $< 10^{-6}$, typically < 10 iterations). Output the raked weights under key `"raked"` in `configs/layer_weights.json` alongside the additive `"additive"` weights. Log the per-iteration change and number of iterations to converge. This is the lightweight middle ground between the additive model and full MRP; it enforces joint consistency across all three marginals without requiring cross-tabulation data. The paper must compare additive vs. raked outputs. [4 hrs]

4 Week 3 — RoBERTa Pipeline, Social Media & Fine-Tuning Dataset (40 hrs)

Goal. By Friday, two parallel sentiment pipelines are running — a news corpus scraper and a four-platform social media collector. Both are scored with RoBERTa, social media sentiment is correlated with 14-day lagged favorability polls, and a unified fine-tuning dataset is assembled.

End condition.

- ☐ `SentimentData` and `SocialMediaSentimentData` artifacts validate and round-trip.
- ☐ At least 400 fine-tuning examples in `train.jsonl`, fusing both news and social media signals.
- ☐ Baseline-adjusted social sentiment computed for all platforms.
- ☐ `eval.jsonl` contains 2020 cycle examples only.
- ☐ PR merged.

Hard fallbacks — read before starting Week 4.

Meta Content Library: If not approved by Week 4 Day 1, do not wait. Activate the fallback immediately: scrape public Facebook page reaction counts (like, angry, sad, love) from major news pages as a valence-only signal. The `FacebookCollector` stub must work in both modes — full Content Library and reaction-count fallback — before Day 2 ends.

Social media collectors: Bluesky and Apify should both be running from Week 0. If Apify's free tier runs out for a given shock, reduce the per-run result count or skip that shock for Apify and rely on Bluesky alone. The historical archives (MeToo hydrated dataset, COVID-19, BLM/George Floyd 2020, 2020 election) are the primary source of Twitter-equivalent signal — live collection is supplementary.

Hailo NPU: If Week 0 compilation failed, confirm `pi_npu.enabled=false` in `configs/base.json`. The CPU fallback path must be running and tested before Day 3's SetFit training begins.

Hard fallback implementation checklist — all three must be complete before the deadlines shown. These are not optional; the pipeline is blocked on them.

- ☐ **[Day 1 end]** `scripts/compile_hailo.py` implemented. The script exports all-MiniLM-L6-v2 to ONNX, runs `hailo optimize` and `hailo compile`, verifies the HEF loads, and writes `pi_npu.enabled=true` to `configs/base.json` on success. On *any* failure it writes `pi_npu.enabled=false` and exits 1. Week 3 is never blocked. **File:** `scripts/compile_hailo.py`
- ☐ **[Day 2 end]** `FacebookCollector` implemented in dual-mode — *not stubs*. `mode="content_library"` uses the Meta Content Library API (requires `FACEBOOK_CL_TOKEN`). `mode="reaction_fallback"` uses the Graph API to scrape public page reaction counts (LIKE, LOVE, WOW, HAHA, ANGRY, SAD) from major news pages and computes a valence score per post. `mode="auto"` (default) auto-selects based on whether `FACEBOOK_CL_TOKEN` is set in the environment. Both modes write the same `posts.jsonl` schema; reaction-fallback records carry a `reaction_valence` field consumed

by `elasticity.py`. **File:** `electoral/nlp/social.py`

- ☐ [Day 2 end] ApifyCollector quota fallback implemented. On actor-run failure, `_call_actor()` returns `None` (not raise). `run()` detects the `None` result, retries once with `max_items=100` (`MAX_ITEMS_QUOTA_RETRY`), and if that also fails, logs a warning and returns 0. The pipeline falls back to Bluesky alone for that shock and continues without crashing. **File:** `electoral/nlp/collectors/apify_x_scraper.py`
- ☐ [Day 3 start — before SetFit training] Hailo CPU fallback verified. Run: `python scripts/pi_bio_server.py --config configs/base.json --cpu-only` and confirm `curl http://pi.local:9000/health` returns `{"status":"ok","mode":"cpu",...}`. The `/health` endpoint returns both `"mode"` (canonical, `"cpu"` or `"npu"`) and `"inference_backend"` (legacy alias). Do **not** start Day 3 SetFit training until this check passes. **File:** `scripts/pi_bio_server.py`

Day 1 (Mon — Shock Taxonomy, API Setup & News Scraper)

☐ **Task:** Build the shock event registry: `configs/shocks.json` listing all historical shock events 2000–2024. Each entry: `id` (slug), `date`, `category` (Security, Geopolitical, Moral/Scandal, Electoral Surprise), `description` (one sentence), `target_blocs`, `keywords`, and `archive_ids` (list of pre-collected dataset slugs that cover this event, or empty list if live collection only). Aim for at least 25 events. Mark the following shocks as archive-covered in `archive_ids`:

- `metoo_2017`: `["metoo_liwc", "reddit_pushshift"]` — Foster & Rathlin 2022 LIWC dataset (3,683 tweets, CC BY-NC-SA 4.0, DOI: 10.5683/SP3/1YWCW1) plus Reddit Pushshift subreddits (`TwoXChromosomes`, `Feminism`, `AskFeminists`, `AskWomen`) filtered to Oct 15–Dec 31, 2017 at sampling time. Shock category: Moral/Scandal. 15 October 2017.
- `kavanaugh_2018`: `["metoo_liwc", "reddit_pushshift"]` — Foster & Rathlin dataset covers `kavanaugh_ford` (27 Sept 2018) and `kavanaugh_confirmed` (6 Oct 2018) phases. Reddit Pushshift filtered to Sept–Oct 2018 at sampling time. Shock category: Moral/Scandal (Electoral). Oct 2018.
- `covid_pandemic_2020`: `["covid_pandemic_2020"]` — ~179k tweets, `gpreda` Kaggle dataset. Shock category: Geopolitical/Health. 11 March 2020, WHO pandemic declaration.
- `covid_vaccine_2020`: `["covid_vaccine_2020"]` — ~200k tweets, `kaushiksuresh147` Kaggle dataset. `user_description` confirmed present — use for bio classification. Shock category: Geopolitical/Health. December 2020, first vaccine approvals.
- Any 2020 cycle events: `["election_2020_ieee", "election_2020_kaggle"]` as applicable
- `blm_george_floyd_2020`: `["reddit_pushshift"]` — **Twitter BLM coverage dropped (2026-06-03)**. Reddit Pushshift `r/BlackLivesMatter` and `r/BlackPeopleTwitter` filtered to May 25–August 31, 2020 is the sole training source. Shock category: Moral/Scandal (Social Movement). Primary signal for African American, secular, and diverse bloc delta bins.

- `daca_rescission_2017`: [] — live collection only; no pre-collected archive. Shock category: Geopolitical (Immigration). Trump rescinds DACA 5 Sept 2017. Target blocs: Latino, Asian, Muslim, Secular.
- `daca_scotus_2020`: ["daca_scotus_2020"] — Schema: `status→text, timestamp→created_at`; no `lang` field (hardcode "en"); no `user.description` (apply Latino/diverse platform proxy, `inference_method="platform_proxy"`). Shock category: Geopolitical (Immigration). 18 June 2020, SCOTUS blocks Trump DACA rescission 5–4 (*DHS v. Regents of the University of California*). Target blocs: Latino, Asian, Secular.
- `election_2016`: ["election_2016"] — kinguistics Kaggle dataset, Election Day tweets only. Shock category: Electoral. 8 November 2016, Trump defeats Clinton. Major shock for Latino, Muslim, Secular, and Jewish blocs. Manually downloaded to `/Volumes/JUHIDRIVE/electoralData/archives/twitter/election_2016/`. **Schema verification required on first load:** confirm `user_description` field is present before assigning final tier. If present: Tier 1 (bio classification). If absent: Tier 2 (platform proxy).
- `election_2012`: ["election_2012"] — jgoodman8 Kaggle dataset, ~1M tweets, JSON format. Shock category: Electoral. 6 November 2012, Obama defeats Romney. Baseline election; lower shock value than 2016 but useful for pre-Trump sentiment calibration. Manually downloaded to `/Volumes/JUHIDRIVE/electoralData/archives/twitter/election_2012/`. **Schema verification required on first load:** confirm `user_description` field is present before assigning final tier. If present: Tier 1 (bio classification). If absent: Tier 2 (platform proxy).
- `election_2024`: ["truth_social_2024"] — USC Humans Lab / kashish-s Kaggle dataset (1.5M posts, Feb 2022–Oct 2024) as primary Truth Social source; plus "x_2024" if sinking8 schema check confirms full objects (not IDs only — still pending). Shock category: Electoral. Covers three sub-events: Trump assassination attempt (13 July 2024), Biden dropout from race (21 July 2024), and Election Day Trump victory (5 November 2024). Target blocs: all five race blocs, Evangelical, Catholic, Secular, women (gender gap). Primary Truth Social source is `truth_social_2024`.

Use the Ayatollah assassination as the pilot/test case throughout this week. [2 hrs]

□ **Task:** Verify all Week 0 social media setup is fully operational before writing any Week 4 code: (i) `BlueSkyCollector` returns posts with author `description` and `langs` fields for the Ayatollah keywords; (ii) `ApifyCollector` returns posts in the correct schema; (iii) the Pi bio server responds at `http://pi.local:9000/health` in either NPU or CPU fallback mode; (iv) Syncthing is syncing between Intel Mac and M5. Log the status of both collectors in `DECISIONS.md` with the date.

Resolution (2026-06-03):

- (i) **BlueSkyCollector** — intentionally deferred. Archive-first strategy adopted; live collectors not needed for historical shock training data. To be configured only if live shock collection is needed post-SRP. Logged in `DECISIONS.md`.

- (ii) **ApifyCollector** — intentionally deferred. Same reason as BlueSky. Logged in DECISIONS.md.
- (iii) **Pi bio server** — **COMPLETE**. Confirmed running at `http://100.125.90.19:9000/health` in CPU fallback mode, returning `{"status":"ok","mode":"cpu","inference_backend":"cpu","model":"all-MiniLM-L6-v2"}`. Hailo NPU compilation skipped. Running from `feature/week4-roberta-social-pipeline` branch.
- (iv) **Syncthing** — **COMPLETE**. Operational and syncing across local machines.

Ayatollah assassination pilot test not run (live collectors deferred). Pilot shock will instead use archive data from the 2020 election dataset at sample time. [1 hrs]

□ **Task:** Compile `all-MiniLM-L6-v2` for the Pi AI HAT+ NPU. On the Pi, run `scripts/compile_hailo.py`: (i) load the model via `sentence-transformers`, (ii) export to ONNX with a dummy input of shape (1,128), (iii) run `hailo optimize minilm.onnx --hw-arch hailo8l`, (iv) run `hailo compile minilm.onnx --hw-arch hailo8l -o models/minilm.hef`. Verify the HEF loads: `from hailo_platform import HEF; HEF("models/minilm.hef")`. This step can run while the Intel Mac setup runs in parallel.

Then start the bio server: `uvicorn pi_bio_server:app --host 0.0.0.0 --port 9000`. **Hostname:** `pi.local` requires mDNS/Bonjour. If it does not resolve from the Intel Mac or M5, assign a static IP to the Pi in your router, add it to `/etc/hosts` on all machines, and update `"pi_bio_server"` in `configs/base.json`. Confirm reachability: `curl http://pi.local:9000/health`.

Resolution (2026-06-03): Hailo NPU compilation **deferred indefinitely**. The Hailo developer SDK is gated behind vendor registration, and the Hailo compiler must run on the host Mac (not the Pi itself) — adding a toolchain dependency that is not justified for the SRP timeline. NPU compilation will only be revisited if CPU inference becomes a bottleneck after SetFit training is complete.

CPU fallback mode is active and sufficient. Bio server confirmed running at `http://100.125.90.19:9000/health` (static IP; mDNS blocked on CMC WiFi). Response: `{"status":"ok","mode":"cpu","inference_backend":"cpu","model":"all-MiniLM-L6-v2"}`. CPU inference (~20 ms/bio on Pi 5) is acceptable for the archive scoring workload. [2 hrs]

→ **Mount JUHIDRIVE before running any pipeline script.**

Drive: /Volumes/JUHIDRIVE — 5 TB exFAT, shared between Mac and Pi over the local network. All raw archives, sampled JSONL, and training data live here; **nothing in archives/ is committed to git.**

Mac path: /Volumes/JUHIDRIVE/electoralData/archives/

Pi path: set "pi_data_root" in configs/base.json

Archive layout on JUHIDRIVE:

- archives/twitter/ — Twitter/X datasets (by shock slug)
- archives/reddit/ — Pushshift subreddit dumps (.zst)
- archives/telegram/ — Telegram message exports
- archives/discord/ — Discord server exports
- archives/snap/ — Snapchat Snap Map data (future)
- archives/validation/ — bot blocklists, held-out sets
- archives/news/ — 3DLNews2 URL metadata + Webhose dumps

Pi network access. The Pi reads and writes archives over the local network via a Samba or NFS mount of JUHIDRIVE. Set "pi_data_root" in configs/base.json to the Pi-side mount path (e.g. /mnt/juhidrive/electoralData/archives/) **before** running any pipeline script on the Pi. This field is separate from "data_root", which is the Mac-side path.

Storage strategy — read before downloading anything.

All raw archives write directly to JUHIDRIVE under /Volumes/JUHIDRIVE/electoralData/archives/{platform}/. Use the platform subdirectory convention: twitter/, reddit/, telegram/, discord/, snap/, validation/, news/.

- **JUHIDRIVE** is the primary archive storage. The 5 TB capacity means raw archives do not need to be deleted after sampling — retain them for schema re-checks and re-sampling runs.
- **3DLNews2:** download only the URL metadata file to JUHIDRIVE archives/news/3dlnews/ (small). Filter by date window and state *before* fetching any HTML. Run the HTML fetch-and-parse as a SLURM array job on HPC scratch so raw HTML is never written to local disk. **rsync** only the extracted `articles.jsonl` back to JUHIDRIVE.
- **Syncthing** syncs processed outputs only (sampled JSONL, cleaned JSONL, scored embeddings). Never configure Syncthing to sync raw archives.
- **M5** holds no raw archives — processed outputs only.

□ **Task:** Download all pre-collected archives and inspect their schemas. Save each to /Volumes/JUHIDRIVE/electoralData/archives/{platform}/{slug}/ and document schema findings in /Volumes/JUHIDRIVE/electoralData/archives/README.md before Day 2.

Tier 1 — full objects, directly usable:

- **MeToo / Kavanaugh LIWC training set (Foster & Rathlin 2022):** DOI: 10.5683/SP3/1YWCW1. Local path: validation/metoo_liwc/fulldataCHBR.sav (2.81 MB, 3,683 tweets). Schema: Tweet→text, Event→shock_slug. No `user_description` — apply secular/diverse platform proxy to all posts. Four shock phases: metoo_2017 (Oct 15, 2017), kavanaugh_ford (Sept 27, 2018), kavanaugh_confirmed (Oct 6, 2018), weinstein_convicted (Feb 24, 2020). **License: CC BY-NC-SA 4.0 — flag for commercial review.** Use as training signal despite small size; supplement with Reddit Pushshift at sampling time. **Note:** The Ruest 25.2M Harvard Dataverse corpus (DOI 10.7910/DVN/2SRSKJ) is **permanently dropped** — do not reference or attempt to download.
- **2020 Election — IEEE (3.5M, July–Aug 2020):** iee-dataport.org/open-access/us-november-2020-election-tweets-dataset.
- **2020 Election — Kaggle (Oct 15–Nov 4 2020):** kaggle.com/manchunhui/us-election-2020-tweets.
- **[DROPPED 2026-06-03] BLM / George Floyd 2020 Twitter (alenjose, Tweet-BLM.csv, BLM2020_Dataset.csv)** — Extensive search found no suitable Twitter/X BLM 2020 dataset with `user_description`: all candidates are IDs-only, wrong language, wrong time period, or too small. The alenjose 100k dataset is entirely 2023 data. Tweet-BLM.csv and BLM2020_Dataset.csv also rejected for the same reasons. Training signal for

blm_george_floyd_2020 comes entirely from Reddit Pushshift r/BlackLivesMatter and r/BlackPeopleTwitter filtered to May 25–August 31, 2020. Delete twitter/blm_2020/ from the archive directory.

- **Congress tweets (ongoing JSON):** alexlitel.github.io/congresstweets/.
- **Polarization corpus (Kiran et al.):** gvrkiran.github.io/polarizationTwitter/. **High priority.** Add bio text + ideological label to data/bio_labels/labeled_bios.jsonl before SetFit training (Week 4 Day 3).
- **USC Telegram 2024 election (leonardo-blas):** github.com/leonardo-blas/usc-tg-24-us-election and huggingface.co/datasets/leonardoblas/us_election_2024_telegram_distilled. No user bio field — apply Evangelical/conservative Protestant platform-level proxy to all posts. Use TelegramLoader.
- **Truth Social — notmooodoo9 (PRIMARY, 31.8M comments + 18k posts):** HuggingFace: notmooodoo9/TrumpsTruthSocialPosts. Downloaded to /Volumes/JUHIDRIVE/electoralData/archives/truthsocial/notmooodoo9/. Slug: truth_social_2025. License: CC-BY 4.0, commercial use permitted.

Three files: truthsocial.comments[Trump-FROM-10-8-25].csv (31.8M rows, ~5.8 GB), truthsocial.posts[Trump-FROM-10-8-25].csv (18,476 Trump posts, Feb 2022–Oct 2025), truthsocial.users[Trump-FROM-10-8-25].csv (1.5M rows — **crawler state table, not profile data**; no bio, description, or display_name field).

Schema (comments): _id (Mastodon snowflake — decode date via `datetime.utcfromtimestamp((id >> 16) / 1000)`), owner (user ID, no bio available), reply_to, text. No created_at column — date must be derived from _id.

Tier 2 — no bio field anywhere in dataset. Apply Evangelical/MAGA/conservative Protestant platform proxy to all comments (`inference_method="platform_proxy"`). Exclude from Σ_{Δ} covariance estimation.

Date range: comments start 8 October 2025 (post-election Trump activity). Trump’s posts go back to Feb 2022; these cover shocks such as trump_indictment_2023, israel_hamas_war_2023, trump_conviction_2024, and trump_assassination_attempt_2024, but as retrospective Truth Social posts rather than reaction-window social media — use for baseline sentiment only, not shock window scoring.

- **Truth Social — Zenodo Notre Dame (truth_social_2022, Feb–Sept 2022, 823K posts):** Downloaded to /Volumes/JUHIDRIVE/electoralData/archives/truthsocial/zenodo_notredame/. Slug: truth_social_2022. Network-structure scrape (12 TSV files): users (454k), truths (854k raw lines, ~824k well-formed), follows, replies, hashtags, media, external URLs. **No bio field** in users.tsv — profile_url is a link, not text. **Tier 2:** apply Evangelical/conservative Protestant platform proxy. Filter `is_retruth=f`; decode dates from Mastodon snowflake _id. Use `csv.QUOTE_NONE` for truths.tsv (embedded newlines in ~30,500 rows).
- **Truth Social — USC Humans Lab / kashish-s Kaggle (PRIMARY 2024**

election source, truth_social.2024): kaggle.com/datasets/kashishashah/truthsocial-2024-election-integrity-initiative. Led by Prof. Emilio Ferrara, USC Humans Lab. 1.5 million posts, February 2022 through October 2024. Save to /Volumes/JUHIDRIVE/electoralData/archives/truthsocial/kashish_usc/. Slug: truth_social.2024.

Shock coverage: covers the full 2024 election window including Trump assassination attempt (13 July 2024), Biden dropout (21 July 2024), and Election Day (5 November 2024). This is the primary Truth Social archive for election_2024 shock analysis.

Tier 2 — no user bio field in Truth Social API data at scrape time. Apply Evangelical/MAGA/conservative Protestant platform proxy (inference_method="platform_proxy"). Exclude from Σ_{Δ} covariance estimation.

Note: github.com/kashish-s/TruthSocial_2024ElectionInitiative is the scraping scripts repository only — it contains no data. The actual dataset is the Kaggle link above.

- **[DROPPED 2026-06-03] BLM / George Floyd 2020 Twitter** — see note above. [twitter/blm_2020/](#) removed from archive structure.
- **COVID-19 tweets (gpreda):** kaggle.com/datasets/gpreda/covid19-tweets. ~179k tweets with #covid19, March–April 2020. Twitter API scrape — confirm user_description field is present on download. Field mapping: text→text, date→created_at, user_description→user.description. Maps to covid_pandemic.2020 shock slug; covers the WHO pandemic declaration shock window (March 11, 2020). Save to /Volumes/JUHIDRIVE/electoralData/archives/twitter/covid_pandemic.2020/.
- **COVID Vaccine tweets (kaushiksuresh147):** kaggle.com/datasets/kaushiksuresh147/covidvaccine-tweets. ~200k tweets with #CovidVaccine, Jan 2020–March 2021. Schema confirmed: user_name, user_location, user_description, user_created, user_followers, user_verified, date, text, hashtags, source, is_retweet. Tier 1 — user_description confirmed present; use directly for bio classification. Maps to covid_vaccine.2020 shock slug; covers vaccine rollout and approval shock window (Dec 2020–March 2021). Save to /Volumes/JUHIDRIVE/electoralData/archives/twitter/covid_vaccine.2020/.

Tier 2 — scorer / demographic training:

- **Political tweets (kaushiksuresh147):** kaggle.com/datasets/kaushiksuresh147/political-tweets.
- **Political partisanship (lingshuhu):** kaggle.com/datasets/lingshuhu/political-partisanship-tweets.
- **Political tweets + gender (lingshuhu):** kaggle.com/datasets/lingshuhu/political-tweets-with-gender-info.
- **PoliticalTweets (HuggingFace):** huggingface.co/datasets/Jacobvs/PoliticalTweets.

- **US Politicians tweets (mrmorj):** kaggle.com/datasets/mrmorj/us-politicians-twitter-dataset.
- **TikTok 2024 election (gabbypinto):** github.com/gabbypinto/US2024PresElectionTikToks. Load caption text as text field; apply younger/secular/diverse proxy. Use TikTokLoader. No bio field available.
- **DACA SCOTUS 2020 tweets (schairezv):** github.com/schairezv/daca-tweets-analysis. Several thousand tweets, June 2020. Covers shock slug `daca_scotus_2020`. Schema: `status`→`text`, `timestamp`→`created_at`; no `lang` field — hardcode "en". No `user.description` — apply Latino/diverse platform proxy to all posts (`inference_method="platform_proxy"`). No `user_id` — bot blocklist cannot be applied; skip deduplication step that depends on author ID. Pre-computed `polarity` and `sentiment` fields are sanity-check reference only — **do not** use as training inputs or fold into $\Delta\mu$ estimation.

ID-only / rehydration backlog — not usable at current API tier:

- **SMAPP NYU elections (SMAPPNYU):** github.com/SMAPPNYU/twitter_elections_public_interest. **Rejected: tweet IDs only.** Rehydration requires X Basic API (\$100/month); not in budget. Same decision as `echen102`, Harvard 2016, and VoterFraud 2020. Revisit if X API pricing changes or a pre-hydrated mirror becomes available.
- **UW ResearchWorks (b88ba40a):** digital.lib.washington.edu/researchworks/items/b88ba40a-4001-4fb4-b960-2ea2e49d58a0. **Rejected: tweet IDs only.** Rehydration requires X Basic API (\$100/month); not in budget. Same decision as SMAPP NYU, `echen102`, Harvard 2016, and VoterFraud 2020. Revisit if X API pricing changes or a pre-hydrated mirror becomes available.
- **X 2024 election (sinking8):** github.com/sinking8/x-24-us-election. Visit repo and check: if full tweet objects, save to `/Volumes/JUHIDRIVE/electoralData/archives/twitter/x_2024/` and add to Tier 1. If IDs only, skip (not usable without paid API).
- **US Pres Elections 2020 (echen102):** github.com/echen102/us-pres-elections-2020. Same check. Emily Chen's datasets are typically IDs — but check for any full-object sample files in the repo.

Bot filter — blocklist only:

- **Political Twitter bots (khanradcoder):** Kaggle: [khanradcoder/political-tweets-from-twitter-bots](https://kaggle.com/khanradcoder/political-tweets-from-twitter-bots). Extract bot user IDs into `/Volumes/JUHIDRIVE/electoralData/archives/validation/bot_blocklist/ids.txt`. **Do not use as training data.**

News archives — historical article text:

- **3DLNews2 (W&M NewsLab) — Tier 1, primary:** github.com/wm-news/3DLNews. 8M+ URLs from 14,000 local outlets, 1995–2024. **Do not fetch HTML yet.** Download the URL metadata file only and save to `/Volumes/JUHIDRIVE/electoralData/archives/news/3dlnews/`. Filter by date window and state before fetching — this reduces the fetch set from millions to tens of thousands

of URLs. Run the HTML fetch-and-parse as a SLURM array job on HPC scratch (see the storage strategy box above). Tag with `source="3dlnews"`.

- **Webhose free news dataset — Tier 2, secondary:** github.com/Webhose/free-news-datasets. Full article text pre-extracted, multi-outlet. Save to `/Volumes/JUHIDRIVE/electoralData/archives/news/webhose/`. Tag with `source="webhose"`.

Reddit archives — Watchful1 Pushshift dump (AcademicTorrents):

- **Source:** Watchful1 Pushshift dump on AcademicTorrents (hash `3e3f64dee22dc304cdd2546254ca1f8e8ae542b4`). Contains the top 40,000 subreddits from 2005-06 to 2025-12 (inclusive) as zstandard-compressed ndjson files. Total torrent is 3.97TB but supports **selective download by subreddit** — only download the identity subreddits needed for bloc proxies.
- **Download only these files:** Catholicism_comments.zst, Catholicism_submissions.zst, BlackPeopleTwitter_comments.zst, BlackPeopleTwitter_submissions.zst, LatinoPeopleTwitter_comments.zst, LatinoPeopleTwitter_submissions.zst, Conservative_comments.zst, Conservative_submissions.zst, Republican_comments.zst, Republican_submissions.zst, politics_comments.zst, politics_submissions.zst, progressive_comments.zst, progressive_submissions.zst, Jewish_comments.zst, Jewish_submissions.zst, Judaism_comments.zst, Judaism_submissions.zst, islam_comments.zst, islam_submissions.zst, Muslim_comments.zst, Muslim_submissions.zst, exchristian_comments.zst, exchristian_submissions.zst.
- **Storage:** save to `/Volumes/JUHIDRIVE/electoralData/archives/reddit/{subreddit_name}/`.
- **Parsing:** use the zstandard library (`pip install zstandard`). Example parsing scripts at github.com/Watchful1/PushshiftDumps.
- **No user bio field on Reddit** — apply subreddit-level demographic proxy per the devplan mapping table. Set `inference_method="subreddit_proxy"` on all posts and exclude from Σ_{Δ} covariance estimation.
- **Single-torrent coverage:** this dump covers all shocks in the registry from 2005 to 2025, so no per-shock Reddit downloads are needed.
- **Additional subreddits — African American bloc:** BlackLivesMatter (direct BLM shock signal), askblackpeople, Blackpeople, BlackPeopleofReddit, blackmen, BlackMentalHealth. **NSFW check required before processing:** BlackIsBetter — inspect content before including; skip if majority off-topic.
- **Additional subreddits — Mormon bloc (new):** mormon, mormonpolitics. Apply Mormon demographic proxy. **Note:** Mormons are a distinct bloc from Evangelical Protestant — they trend Republican but broke partially from Trump in 2016 and 2020. Save to `/Volumes/JUHIDRIVE/electoralData/archives/reddit/mormon/`.
- **Additional subreddits — Secular/progressive bloc:** Liberal, AskALiberal, LibertarianLeft, ProgressiveHQ, atheism.

- **Additional subreddits — Deconversion signal:** `exl1dm` (ex La Luz del Mundo — Latino religious deconversion signal), `exmormon` (now complete with Mormon additions above).
- **Do not process — accidental downloads (irrelevant):** `askajudge`, `askanelectrician`, `aternos`, `atheismbot`, `atheismindia`, `ateismo.br`, `morganbailley`, `morningsomewhere`, `moroccancuties1`, `morrissey`, `mortarsportfeed`, `mossberg`, `mostre_os_peitos`, `mother4`, `motherbusnark`, `RefugeOfSilksong`, `RegalUnlimited`, `Referrals`, `atc2`. Delete these files after confirming they are not needed.
- **Incomplete downloads — re-download from torrent:** `politics_comments.zst`, `Conservative_submissions.zst`, `SoCallLatina_comments.zst`, `DebateACatholic_submissions.zst`, `excatholic_comments.zst`. Verify file size against the torrent manifest before processing.

For each dataset, confirm presence of `text` (or `full_text`), `created_at`, and `lang`. Record all findings in the archives README before Day 2.

Resolution (2026-06-03): Schema verified for all downloaded archives. Full findings in `/Volumes/JUHDRIIVE/electoralData/archives/README.md`. Summary:

- **election_2012 (tweets.json):** Tier 1. Fields: `text`, `created_at` (Twitter date string), `user.description` confirmed present. **No lang field** — hardcode "en".
- **election_2016 (election_day_tweets.csv):** Tier 1. Fields: `text`, `created_at`, `lang`, `user.description` (dot-notation CSV column) — all confirmed.
- **election_2020 (manchunhui Kaggle):** Tier 1. Two files (`hashtag_donaldtrump.csv`, `hashtag_joe Biden.csv`). Fields: `tweet`→`text`, `created_at`, `user.description` confirmed present. **No lang field** — infer with `langdetect` or hardcode "en". Deduplicate by `tweet_id` across both files before sampling.
- **x_2024 (sinking8 / USC):** Tier 1 confirmed — full objects, NOT IDs-only. CSV.gz, `snsrape` format, ~47 parts × ~1M tweets each. Fields: `rawContent`→`text`, `date`→`created_at`, `lang`, `user.rawDescription`→`bio` (confirmed populated). **Note:** `user` column is a Python dict repr — parse with `ast.literal_eval()` to extract `rawDescription`. Filter by date window per shock slug.
- **kashish_usc (truth_social_2024):** Tier 2. 1,585,187 rows confirmed. Fields: `status`→`text`, `timestamp`→`created_at`, `author_username`. No bio, no lang column. Platform proxy: Evangelical/MAGA/conservative Protestant.
- **telegram_2024:** Tier 2. Fields: `content`→`text`, `date`→`created_at`, `language`→`lang`, `from_id`→`author`. Platform proxy: Evangelical/conservative Protestant.
- **TikTok (tiktok/):** Tier 2 with Whisper transcripts. `TikTok.IDs_and_Transcripts_Published_2025_Feb.csv` has single column `whisper_voice_to_text` (~53% coverage). No `created_at`, no `lang`, no bio. Use non-empty transcript rows only; hardcode "en".
- **blm_2020 (alenjose):** **DROPPED**. All 100,000 tweets are from 2023-04-03, not May–June 2020. `TweetBLM.csv` and `BLM2020_Dataset.csv` also rejected — same issues

(IDs-only, wrong language, or wrong time period). **Decision (2026-06-03):** BLM 2020 training signal comes entirely from Reddit Pushshift `r/BlackLivesMatter` and `r/BlackPeopleTwitter` filtered to May 25–August 31, 2020. `twitter/blm.2020/` removed from active archive structure.

- **Reddit Pushshift:** all targeted subreddits downloaded. `DebateAChristian_comments.zst` and `DebateCommunism_comments.zst` are `.part` files — re-download needed. `politics_comments.zst` (26 GB) and `Conservative_submissions.zst` (410 MB) are full-size and assumed complete.
- **3DLNews2:** still downloading as of 2026-06-03. Do not attempt HTML fetch until download completes.

[5 hrs]

□ **Task:** Write `scripts/sample_archives.py` and submit it as a **HPC SLURM array job** to draw stratified samples from the raw archives. This runs *before* any RoBERTa scoring and reduces the corpus from hundreds of millions of posts to a tractable, representative sample.

Three strata per archive per shock event:

1. **Temporal:** sample proportionally by day across the shock window (e.g. 3 days before, day-of, 3 days after, 7 days after) so the full arc of public reaction is captured rather than just the peak day.
2. **Demographic:** after a fast keyword-based bio pass, sample proportionally by inferred bloc so all 10 groups are represented. Without this step, high-volume blocs (secular, evangelical) crowd out low-volume blocs (Jewish, Muslim).
3. **Sentiment extremity:** run a lightweight keyword sentiment score (VADER or similar) and oversample posts at the extremes (top and bottom 20% of the distribution). Extreme examples are more informative for fine-tuning than neutral ones.

Target output: ~200,000 posts per major archive dataset, ~5,000 per shock event per platform. Submit as a SLURM array job (`#SBATCH --array=0-N` where N is the number of archive/shock combinations). Add `just sample` to the Justfile.

Status (2026-06-09): Sampling complete on USC Laguna HPC — **1,553,246 posts** across **77 shock/archive combinations**, submitted as SLURM array job (`scripts/hpc/sample_archives_laguna.slurm`). All three strata implemented: temporal (proportional by day across shock window), demographic (keyword bio pass with religion→race→gender priority, CNN 2024 NEP weights), sentiment extremity (VADER $2 \times$ weight on $|\text{compound}| \geq 0.5$). Target of 200,000 posts per major archive met for `election_2012` (200,000), `election_2020` (113,659), `blm_george_floyd_2020` (74,938). Reddit Pushshift covers 2013–2025 shocks; `reddit_monthly` covers pre-2013 shocks pending `filter_reddit_monthly` completion. 3DLNews local news parsed for 16 states, 2015–2024. `just sample` and `just sample-laguna` added to Justfile.

Two pending items before fully complete:

1. `filter_reddit_monthly_laguna.slurm` submitted but output not yet sampled — will add ~200k posts for pre-2013 shocks once filtering completes.
2. Discord extraction jobs on Laguna not yet run.

[3 hrs]

□ **Task:** Write `scripts/clean_with_llm.py` and run it on the **M5** using a local open-weight model (Mistral-7B-Instruct) (free tier: 1,000 requests/day) during the initial build phase. In continuous mode post-SRP, this script runs on the **Intel Mac** as part of the nightly launched pipeline. Four cleaning operations:

1. **Off-topic detection.** For each post, prompt Gemini: *“Is this post about [shock event]? Reply yes or no.”* Drop all “no” responses. This removes posts that mention the shock keyword coincidentally (e.g. MeToo off-topic tweets such as entertainment industry gossip unrelated to political activation).
2. **Spam and low-information filtering.** Remove posts that survive the bot blocklist but add no signal: pure retweet text, copy-paste chain tweets, emoji-only posts, bare URL shares with no commentary.
3. **Text normalisation.** Expand political abbreviations (SCOTUS, POTUS, MAGA, FLOTUS), normalise hashtag formatting, strip URLs and @mentions so RoBERTa receives cleaner input without domain-specific noise.
4. **Cross-platform deduplication.** The same post sometimes appears in multiple archives. Deduplicate on `hash(user_id + text[:50])` before scoring.

Throughput: ~50 posts per Gemini prompt $\times$ 1,000 requests/day = 50,000 posts/day. For 200k posts per archive this takes 4 days. **Start the most important archives (COVID pandemic tweets from Kaggle (gpreda), COVID vaccine tweets from Kaggle (kaushiksuresh147), Reddit BLM 2020, 2020 election) during Weeks 2–3** while other components are being built, so cleaned data is ready for the Week 4 RoBERTa scorer. Add just `clean` to the Justfile. Log the discard rate per cleaning step in `/Volumes/JUHIDRIVE/electoralData/archives/README.md`.

Status (2026-06-09): Cleaning complete — **1,553,246** sampled posts $\rightarrow$ **231,725** cleaned posts (**14.9%** kept) across **77 shock/archive combinations**. Off-topic filter via Gemini 2.5 Flash Lite with async concurrency (50 concurrent batches, `asyncio.gather`). Discard breakdown: ~85% off-topic, <1% spam, <1% dedup. Cleaned data at `/scratch/JDamley28@cmc.edu/electoralData/cleaned/` on Laguna and synced to `/Volumes/JUHIDRIVE/electoralData/cleaned/`. [3 hrs]

□ **Task:** Implement `electoral/nlp/scrapper.py` on the **Intel Mac**: `scrape_articles(shock_id, sources, date_range)` for news sources (Christianity Today, CBN, Univision, NYT, WaPo, Fox). Strip HTML, extract plain text, save to the `Synthing rawdata/articles/` folder so the M5 can read results immediately. Enforce 100-word minimum, log discard rate, test on Ayatollah shock.

Status (2026-06-10): Implemented as `electoral/nlp/scrapper.py` using direct RSS feeds

(BBC, Guardian, NPR, Christianity Today, CBN, Fox News). Google News RSS dropped due to URL encoding issues. Tested on `iran_war_2026` shock—4 articles written from BBC and Guardian. Scraper is live-collection only; RSS feeds do not archive past articles so historical shocks cannot be scraped. `just scrape` added to Justfile. [1 hrs]

□ **Task:** Implement `electoral/nlp/news_loader.py`: `NewsArchiveLoader` that normalises pre-collected news articles into the same `articles.jsonl` schema as the live scraper, so the RoBERTa scorer is blind to source.

Two sources, handled as subclasses or branches:

- **3DLNewsLoader:** reads URL entries from `/Volumes/JUHIDRIVE/electoralData/archives/news/3dlnews/`; filters to the shock date window using `publication_date`; fetches and HTML-strips each article (same logic as `scraper.py`); optionally filters by `state` or outlet name for targeted demographic coverage; enforces 100-word minimum; tags `source="3dlnews"`. This is the primary historical news source — 14,000 local outlets covering the communities that constitute each of the 10 blocs.
- **WebhoseLoader:** reads pre-extracted text from `/Volumes/JUHIDRIVE/electoralData/archives/news/webhose/`; filters by `published` date and optionally by `url` domain; enforces 100-word minimum; tags `source="webhose"`. No HTML fetch needed.

Both write to `rawdata/articles/{shock_id}/` with fields `{id, text, published, url, source, archive_id}`. Test `3DLNewsLoader` on the `metoo_2017_2023` shock, filtering to Oct–Nov 2017 articles from Pennsylvania, Georgia, and Texas local outlets as a demographic cross-section.

Status (2026-06-10): Implemented as `electoral/nlp/news_loader.py` with `ScrapedNewsLoader`, `3DLNewsLoader`, and `WebhoseLoader`. Tested `3DLNewsLoader` on `metoo_2017` shock filtered to PA, GA, TX states Oct–Nov 2017: loaded 16,095 articles (51,303 skipped by date, 55,853 by state, 4,794 under 100 words). Outlet-level demographic proxies embedded in `author_description`. Module is a library—imported by `score_roberta.py` and other pipeline scripts. [2 hrs]

Day 2 (Tue — Social Media Collectors & Bio Keyword Lexicon)

□ **Task:** Implement `electoral/nlp/social.py` with a `SocialCollector` abstract base class and the following concrete subclasses. All write identically-formatted `posts.jsonl`; the scorer is blind to source.

- **BlueSkyCollector:** primary free live path. AT Protocol `app.bsky.feed.searchPosts` via `atproto`. Returns `author_description` and `language` fields for bio classification. No rate limit on public posts.
- **ApifyCollector:** secondary free live path. `apify-client` Python SDK running the X Scraper actor. Covers the Evangelical, Catholic, and urban demographic profile that Bluesky underrepresents. Free tier: 500 results/run. **Do not implement XCollector or install tweepy — the X Basic API is not used.**

- **TruthSocialCollector:** Mastodon-compatible `/api/v1/timelines/public?q=` endpoint. No auth. Small user base; weight accordingly.
- **FacebookCollector:** Meta Content Library client if approved, else public page reaction-count fallback.

For the Ayatollah pilot: collect at least 1,000 posts from Bluesky and run Apify for the same event to confirm schema parity.

Status (2026-06-10): Implemented as `electoral/nlp/social.py` with `SocialCollector` ABC and `BlueSkyCollector`, `ApifyCollector`, `TruthSocialCollector`, `FacebookCollector` subclasses. `BlueSkyCollector` tested on `ayatollah_assassination` and `iran_war_2026` shocks—317 posts in 60 seconds, schema confirmed correct. `author_description` null until Pi bio server classifies bios in scoring stage. Apify schema parity test pending Apify token configuration. TruthSocial and Facebook collectors implemented but untested. [3 hrs]

□ **Task:** Implement `electoral/nlp/archive.py`: `HistoricalArchiveLoader` class with `load(shock_id, archive_id, window_hours=72, bot_blocklist=None)`. Core behaviour: (i) reads raw archive files from `/Volumes/JUHIDRIVE/electoralData/archives/{platform}/{archive_id}/`; (ii) filters to the shock date window; (iii) drops posts whose `user.id` is in `bot_blocklist` and logs the count; (iv) normalises to `{id, text, created_at, lang, bloc_weights, source, archive_id}`; (v) calls `/classify_batch` for posts with `user.description`; (vi) applies platform proxy for posts without a bio field; (vii) saves to `rawdata/social/archive/{shock_id}/`.

Implement two additional loaders as subclasses or standalone functions:

- **TelegramLoader:** normalises Telegram message objects (field names differ from Twitter — `message` instead of `text`, `date` instead of `created_at`, no `user.description`). Applies Evangelical/conservative Protestant as the platform-level proxy for all posts. Tags with `platform="telegram"`.
- **TikTokLoader:** normalises TikTok video metadata (caption text as `text`, `createTime` as `created_at`, no bio field). Applies younger/secular/diverse as the platform-level proxy. Tags with `platform="tiktok"`.

Handle Twitter schema differences: the MeToo dataset uses standard Twitter v2 JSON `text` field; normalise all legacy archives to `text`. Test the main loader on `metoo_2017_2023` filtered to Oct 15, 2017 ± 14 days (MeToo origin week). Test `TelegramLoader` on the USC 2024 dataset. Verify all outputs match the `posts.jsonl` schema exactly.

Status (2026-06-10): Implemented as `electoral/nlp/archive.py`. `HistoricalArchiveLoader` tested on `metoo_2017/metoo_liwc`—SAV support added, loads correctly. `TelegramLoader` tested on `election_2024/telegram_2024`—3,122 posts loaded from CSV format. `TikTokLoader` implemented but untested (no TikTok data available). All outputs match `posts.jsonl` schema. [4 hrs]

□ **Task:** Build the keyword lexicon in `electoral/nlp/bio_classifier.py`: a `BIO_LEXICON` dict mapping each canonical `BlocId` to a list of bio keyword patterns. Source the patterns

from Prof. Espinosa’s published research on linguistic and cultural markers per bloc. Required entries per bloc:

- **african_american**: “Black”, “African American”, “HBCU”, “#BlackTwitter”, “NAACP”
- **latino**: “Latino”, “Latina”, “Latinx”, “Chicano”, “Hispanic”, bio in Spanish
- **asian**: “Asian American”, “AAPI”, bio in Chinese/Korean/ Vietnamese/Tagalog/Hindi
- **secular**: “atheist”, “agnostic”, “humanist”, “no religion”, “nonreligious”
- **jewish**: “Jewish”, “#JewishTwitter”, “synagogue”, “rabbi”, bio in Hebrew
- **muslim**: “Muslim”, “Islam”, “Alhamdulillah”, “Inshallah”, “mosque”, bio in Arabic
- **catholic**: “Catholic”, “parish”, “rosary”, “Vatican”, “#CatholicTwitter”
- **protestant**: “Protestant”, “Lutheran”, “Methodist”, “Presbyterian”, “Baptist” (non-Evangelical)
- **evangelical**: “Evangelical”, “born again”, “Southern Baptist”, “#ChristianTwitter”, “Bible-believing”

The bio classifier outputs **three separate probability distributions** per post (one per stratum): **race_weights** over {african_american, latino, asian, white, other_race}, **religion_weights** over {evangelical, catholic, protestant, secular, jewish, muslim, other_rel}, and **gender_weights** over {women, men, other_gender}. These are three independent distributions; no cross-tabulation is produced.

Gender signal (**gender_signal**: "F" | "M" | null) is extracted as a third field from pronouns (“she/her”, “he/him”), explicit words (“woman/women”, “man/men”, “mom”, “feminist”), and name-based inference as a last resort.

Store as: `configs/race_lexicon.json`, `configs/religion_lexicon.json`, `configs/gender_lexicon.json`, all loaded at runtime. [3 hrs]

□ **Task: Language fallback — validation-only, exclude from all estimation.** When both keyword matching and SetFit return no signal (bio empty or below confidence threshold), fall back to language-based Pew priors (e.g. “es” → {latino/catholic: 0.38, latino/protestant: 0.22, ...}). Tag every post assigned via this fallback with **inference_method**: “language_prior” in the post record.

Critical: do *not* include language-prior posts in mean μ or covariance Σ_{Δ} estimation. Including them in mean estimation while excluding from covariance bifurcates the data generating process — μ and Σ would come from different corpora, violating the mean-variance optimizer’s assumptions. Use language-prior posts only as a held-out validation set: after estimation, compare their sentiment signal against the bio-classified signal and report any divergence in the paper. Store language priors in `configs/language_priors.json`. [2 hrs]

□ **Task:** Implement `TruthSocialCollector` and `FacebookCollector` in `social.py` (**not stubs** — see hard fallback checklist above), and `TelegramLoader` and `TikTokLoader` in

archive.py. TruthSocialCollector uses the Mastodon-compatible /api/v2/search endpoint at truthsocial.com (no auth). FacebookCollector must support mode="content_library" and mode="reaction_fallback" before this task is marked done. Truth Social, Facebook, Telegram, and TikTok posts do not provide structured author bio fields — for all of these apply the platform-level demographic proxy directly (inference_method="platform_proxy").

Status (2026-06-10): All four classes implemented. TruthSocialCollector tested—returns 403 Forbidden on unauthenticated requests (Truth Social has tightened API access since devplan was written). Fails gracefully with 0 posts returned, no crash. FacebookCollector implemented with both content_library and reaction_fallback modes. TelegramLoader tested and working (3,122 posts for election_2024). TikTokLoader implemented but no TikTok data available to test. inference_method=platform_proxy applied correctly to all platforms without bio fields. [3 hrs]

Day 3 (Wed — Bio Classifier: Embedding Model, Language Signal & Scorer)

□ **Task:** Build the manual bio labelling dataset: create data/bio_labels/labeled_bios.jsonl with 100–200 real user bios per bloc (aim for 1,000+ total). Source bios from the polarization corpus (Kiran et al., already downloaded in Day 1 — ideologically-sorted user bios with labels), the Kaggle political tweets datasets, and the senator-tweets corpus (m-newhauser/senator-tweets, already downloaded). For any blocs underrepresented in the archives, manually write example bios from public Bluesky profiles. Bios matching the keyword lexicon from Day 2 can be auto-labelled; focus manual effort on ambiguous cases.

Status (2026-06-10): data/bio_labels/labeled_bios.jsonl created with 2,191 labeled bios. Distribution: 150 per race bloc (white, latino, african_american, asian, other_race), 150 per religion bloc (catholic, protestant, secular, muslim, other_rel, jewish, evangelical), 150 per gender signal (M, F). All labeled via keyword_auto from election_2020 and other corpus bios. No manually labeled ambiguous cases yet—acceptable for SetFit pre-training since keyword-matched bios are high confidence. Manual labeling of ambiguous cases deferred to post-SRP phase. [2 hrs]

□ **Task:** Train the SetFit bio classifier (Option B): install setfit. Load all-MiniLM-L6-v2 as the embedding backbone. Run SetFitTrainer(model, train_dataset, eval_dataset) with the labelled bios; target > 0.75 macro-F1 across all 10 blocs. Export the trained head weights.

Update scripts/pi_bio_server.py to handle both execution modes explicitly. On startup, read configs/base.json["pi_npu_enabled"]:

- **If true:** load the embedding backbone from the compiled HEF via the Hailo SDK. SetFit classification head runs on CPU as a second step.
- **If false:** load all-MiniLM-L6-v2 via HuggingFace SentenceTransformer into CPU memory. Do not attempt to initialise the Hailo SDK or load any HEF. The SetFit classification head runs on CPU as normal.

Never attempt to pass data to the Hailo interface when NPU is disabled — this causes an immediate crash. The `/health` endpoint must return a JSON object with `"status": "ok"` and `"mode"` set to either `"npu"` or `"cpu"`, so callers can confirm which path is active. Test both modes explicitly before Week 4 Day 3 begins. [3 hrs]

Status (2026-06-07): server infrastructure and NPU/CPU path separation complete (scripts/pi_bio_server.py rewritten). Training script written (scripts/train_setfit.py). 28 bio-classifier tests pass. Prerequisite blocking actual model training: data/bio_labels/labeled_bios.jsonl does not yet exist — run build_bio_labels.py on M5 with JUHIDRIVE mounted to generate it, then train_setfit.py. macro-F1 target of 0.75 not yet measured (no trained weights exist).

□ **Task:** Implement `electoral/nlp/scorer.py` with **embedding caching** (Option H): `score_text(text)` first checks `data/embeddings/{post_id}.parquet` for a cached embedding vector. If found, skip the model forward pass and use the cached vector directly. If not found, run the model, save the embedding to Parquet, then score. Implement `score_posts_weighted(posts)` using the SetFit bio weights. The first run of the scorer is slow (embeds everything); all subsequent runs with different aggregation logic are near-instant. Log cache hit rate. [3 hrs]

Day 4 (Thu — Lagged Favorability Correlation & Elasticity Regression)

□ **Task:** Generate **synthetic training data** using Gemini 2.5 Pro (Option D). Run `scripts/generate_synthetic.py`: (i) construct a single prompt containing the full voter panel (as CSV), all 25+ real shock records, the canonical stratum IDs, and instructions to generate 500–1,000 plausible hypothetical shock scenarios with demographically grounded delta estimates per stratum; (ii) parse the JSON response into JSONL records matching the fine-tuning schema; (iii) mark each record `"synthetic": true` and assign an `mmd_weight` per the formula below; (iv) save to `data/finetune/synthetic.jsonl` and diagnostics to `data/finetune/synthetic_diagnostics.json`.

Three required diagnostics (compute for every batch; store in diagnostics file):

- **MMD:** compute RBF-kernel MMD between synthetic and real joint $\Delta\mu$ distributions. Set $\lambda = 1/(2\sigma^2)$ where $\sigma^2 = \text{median pairwise distance over the real data (median heuristic)}$. Assign weight $w = 0.5 \times \exp(-\lambda \cdot \text{MMD}^2)$.
- **PCA alignment:** fit PCA on real $\Delta\mu$ matrix; project synthetic batch onto the top-2 PCs; flag if $< 70\%$ of synthetic variance is explained.
- **PCD (Pairwise Correlation Difference):** compute Frobenius norm of (real pairwise correlation matrix – synthetic pairwise correlation matrix) normalized by number of off-diagonal elements. Log for paper.

Never use a hard KS-test rejection gate — the soft MMD weight is sufficient.

Status (2026-06-08): scripts/generate_synthetic.py complete. Generated 557 synthetic shock scenarios via Gemini 2.5 Pro covering all 15 blocs with delta_bin estimates. Saved to

`data/finetune/synthetic.jsonl`. Diagnostics (MMD, PCA alignment, PCD) deferred—require ≥ 3 real scored shocks in `data/finetune/real.jsonl`. Will recompute after RoBERTa scorer runs on cleaned corpus. [3 hrs]

□ **Task:** Build `configs/market_contracts.json`: a static mapping of every shock event to its exact contract identifier on each supported prediction market platform. Querying Polymarket, Manifold, or Metaculus with a generic slug like `metoo_2017_2023` returns empty or irrelevant results — each platform uses its own internal IDs that must be looked up manually. Each entry maps a shock slug to a dict of `{platform: contract_id_or_null}`, for example: `"metoo_2017_2023": {"polymarket": null, "predictit": null, "metaculus": "3712", "manifold": null, "iem": null}`. For each of the 25+ shock events, visit each platform’s search interface, find the closest relevant contract, and record the exact ID. Mark `null` where no contract exists. This file is the source of truth for the market collector — no dynamic slug resolution. Commit to the repo and add to the replication package. [2 hrs]

□ **Task:** Implement **prediction market collection** in `electoral/markets/collector.py`. Load contract identifiers from `configs/market_contracts.json` (built in the previous task) — never resolve slugs dynamically; use only the hardcoded IDs. For each of the 25+ shock events, collect the market-implied win probability for both the Democrat and Republican candidate contracts at $t - 24h$, $t + 1h$, $t + 24h$, and $t + 72h$. API endpoints:

- **Polymarket** (2020+): `clob-api.polymarket.com` REST API. Free, no auth required. Query by CLOB market address from `market_contracts.json`.
- **PredictIt** (2014+): `www.predictit.org/api/marketdata/all/` JSON feed. Free, no auth. Query by contract ID from `market_contracts.json`. Download full historical CSV from predictit.org/research.
- **Metaculus** (2015+): `www.metaculus.com/api2/questions/` Free REST API. Query by question ID from `market_contracts.json`.
- **Manifold** (2022+): `manifold.markets/api/v0/markets` Free REST API. Query by market slug from `market_contracts.json`.
- **Iowa Electronic Markets** (1988+): Download historical CSV from tippie.uiowa.edu/iem. Parse offline. Best coverage for 2000–2014.

Implement `electoral/markets/aggregator.py`: `volume_weighted_price(market_prices, volumes)` that weights each market’s price by its trading volume; assign equal weight when volume is unavailable (Metaculus, Manifold). Construct a `PredictionMarketData` artifact per shock event and write to `artifacts/markets/{shock_id}.json`. For events before 2014, set `prediction_market = null` in the fine-tuning record. [3 hrs]

□ **Task:** Implement `electoral/nlp/elasticity.py`: `fit_elasticity(news_scores, social_scores, polling_deltas)` that regresses observed vote-share deltas on both news RoBERTa scores and baseline-adjusted social media scores, using platform as a demographic proxy weight. Use `sklearn.linear_model.Ridge` with cross-validation on held-out shock events. Store coefficients per source (news vs. each platform vs. archive) per bloc. For shocks

with archive coverage, the archive posts contribute to the same `social_scores` dict as live posts — they are already normalised to the same schema and the scorer has processed them identically. [3 hrs]

□ **Task:** Compute and log the per-platform correlation between baseline-adjusted social sentiment and 14-day lagged favorability deltas. Write results to `docs/platform_correlations.md`. If a platform shows near-zero correlation, down-weight or exclude it in the regression — document the decision. [2 hrs]

Day 5 (Fri — Merge Stage, Dataset Assembly, Full Corpus Run & PR)

□ **Task:** Implement `merge_posts(shock_id)` in `electoral/kernels/sentiment.py` as a Prefect task. This stage runs *before* the RoBERTa scorer and is required because multiple files may exist per shock (one per platform per collector run). The task: (i) finds all `rawdata/social/{platform}/{shock_id}/.jsonl` files across all platforms and all archive sources; (ii) concatenates them into a single `rawdata/merged/{shock_id}/posts.jsonl`; (iii) logs the total post count broken down by platform so you can see the coverage split. The scorer always reads from `rawdata/merged/` and is completely blind to how many files or platforms contributed. Wire this task into `pipeline.py` as the stage immediately preceding `score_posts`. Verify: running the pipeline twice produces identical merged files. [2 hrs]

□ **Task:** Implement `build_finetune_dataset` in `elasticity.py`: create one JSONL record per shock event using the unified schema. The `social_roberta_scores` field is keyed by `bloc_id`; include the `prediction_market` block from the `PredictionMarketData` artifact (set to null for pre-2014 events). Apply differential weighting: market-backed examples get `weight=1.5`, poll-only examples get `weight=1.0`, synthetic examples get `weight=0.5`. Write to `data/finetune/train.jsonl` (all cycles except 2020) and `data/finetune/eval.jsonl` (2020 only). Validate: no duplicate shock IDs, all 10 bloc keys present, no train/eval overlap. [3 hrs]

□ **Task:** Submit the RoBERTa scoring pipeline as a **USC Laguna SLURM array job** across all 25+ shock events simultaneously.

First: confirm sampled and cleaned data is in place. The scorer reads from `rawdata/merged/{shock_id}/posts.jsonl` which should contain the output of `sample_archives.py` and `clean_with_llm.py`, not the raw full-corpus files. Confirm post counts are in the 5,000–200,000 range per shock, not millions. If raw data was accidentally staged, re-run sampling first.

Then: transfer and submit. Do not rely on Synthing on USC Laguna nodes — scratch nodes kill background daemons. Run:

- `rsync -avz rawdata/merged/ laguna:$SCRATCH/electoral/merged/`
- `rsync -avz rawdata/articles/ laguna:$SCRATCH/electoral/articles/`

Wait for `rsync` to complete before submitting the job.

Then: write and submit `scripts/score_array.sh`: `#SBATCH --array=0-24`, one node per shock event, each node loads its assigned `posts.jsonl` from `$SCRATCH/electoral/merged/` and runs the weighted scorer. Collecting 25 events serially on the M5 would take 2–3 days; the array job finishes in under 2 hours on USC Laguna. Monitor with `squeue -u $USER`.

Scorer sanity check — LIWC validation (run after scoring completes): `validation/metoo_liwc/fulldataCHBR.sav` (Foster & Rathlin 2022, DOI: 10.5683/SP3/1YWCW1) contains 3,683 LIWC-pre-scored tweets across four shock phases. After the RoBERTa scorer has processed the MeToo/Kavanaugh archives, load this file and compute the Spearman correlation between RoBERTa emotion outputs and the LIWC Tone, posemo, negemo, and anger columns on the same tweet IDs. **If Spearman $\rho < 0.4$ on negemo or anger, do not proceed to SetFit training.** Flag the result in `DECISIONS.md` (column name, ρ value, and the decision) before moving to Day 3. **[3 hrs]**

☐ **Task:** Write `tests/test_bio_classifier.py`: (i) keyword match for an explicit Evangelical bio returns `evangelical` weight > 0.8 , (ii) Spanish-language `lang` code boosts `latino` weight, (iii) `combine_signals` output sums to 1.0 for all inputs, (iv) unclassifiable bio returns platform fallback. Write `tests/test_social.py`: (i) `BlueSkyCollector` records include `author.description` and `lang` fields, (ii) `ApifyCollector` output schema matches `BlueSkyCollector` exactly, (iii) baseline adjustment produces per-bloc scores in $[-1, 1]$, (iv) weighted scorer produces 10 distinct elasticity values not a single platform score. Write `tests/test_sentiment.py`: (i) scorer outputs sum to 1.0, (ii) dataset has no train/eval overlap. **[1 hrs]**

☐ **Task:** Weekly PR: `feature/week4-roberta-social-pipeline`. Include: dataset statistics (examples, shock categories, mean elasticity per bloc), platform correlation table, and the Ayatollah pilot findings. **[1 hrs]**

☐ **Task:** Brief supervision check-in with Prof. Espinosa: share platform correlation results and elasticity scores for 3–4 key shocks. Get domain input on whether Truth Social’s Evangelical reactions align with his published patterns. Juhi decides on platform weighting adjustments and documents in `DECISIONS.md`. **[2 hrs]**

5 Week 4 — LLM Fine-Tuning & Inference Endpoint (40 hrs)

Goal. By Friday, Mistral 7B is fine-tuned via QLoRA on the unified news + social media dataset on USC Laguna, the FastAPI endpoint returns valid `ShockResponseData` JSON with constrained decoding, and the three-way baseline comparison (news-only, social-only, unified) is documented.

End condition.

- ☐ 100 consecutive inference calls all return schema-valid JSON.
- ☐ Endpoint responds in under 3 seconds on M5.
- ☐ Three-way MAE comparison (news-only, social-only, unified) committed.
- ☐ PR merged.

Day 1 (Mon — Schema Migration, USC Laguna Setup & Local MLX Baseline)

☐ **Task: Migrate ShockResponseData to the per-stratum schema before writing any LLM inference code.** The current `artifacts.py` stores a flat `deltas: dict[str, float]` over all 15 blocs with an $N \times N$ covariance ($N = 15$). This contradicts two architectural decisions in `DECISIONS.md`: (i) the 5×5 race-only covariance rule (religion and gender do not enter Σ_Δ), and (ii) the requirement to persist the raw LLM 9-token bin output alongside the converted floats.

Replace the current fields with:

- `delta_bins_race, delta_bins_religion, delta_bins_gender: dict[str, str]` storing the raw constrained-decoding token (one of the nine bin labels) per bloc. Required for fine-tuning reproducibility.
- `deltas_race, deltas_religion, deltas_gender: dict[str, float]` storing the converted $\Delta\mu$ per bloc.
- `delta_eff: float` — scalar effective delta $\lambda_1 \sum w_i \Delta\mu_i^{\text{race}} + \dots$
- `covariance: list[list[float]]` — 5×5 race-bloc only. Validation: `assert side == 5`, not N .

Update `validate()` to assert the 5×5 covariance shape and validate all three `delta_bins.*` dicts against the nine canonical bin labels. Update `build_shock_response` in `stages.py` and its tests in `test_stages.py` to match the new schema. Do *not* start USC Laguna setup until this migration is committed and all tests pass — the LLM endpoint return type depends on this schema. [2 hrs]

☐ **Task:** Set up USC Laguna access: log in via SSH, confirm SLURM quota and available GPU nodes (`sinfo`), check scratch storage allocation (`quota`). Create the project directory at `$SCRATCH/electoral-equilibrium/`. Document node names, partition names, and max walltime in `scripts/LAGUNA_NOTES.md`. *Note: USC Laguna is used for all training and compute; CMC HPC is reserved for hosting (vLLM deployment) only.* [1 hrs]

□ **Task:** Copy the fine-tuning dataset to USC Laguna scratch: `scp -r data/finetune/laguna:$SCRATCH/electoral-equilibrium/data/`. Verify file counts match locally. Set up a Python virtual environment on Laguna: `python -m venv $SCRATCH/venv` and install `transformers`, `peft`, `bitsandbytes`, `torch`, `accelerate`. Confirm `import torch; torch.cuda.is_available()` returns `True` on a GPU node. [2 hrs]

□ **Task:** Write `scripts/laguna_submit.sh`: a SLURM batch script for USC Laguna that requests 1 GPU (A100 if available, else V100), 32GB RAM, 4 CPU cores, and 4-hour walltime. The script activates the `venv`, sets `TRANSFORMERS_CACHE` to scratch, and calls `python -m electoral.llm.trainer --backend hpc --config configs/train_r16.json`. Test with `sbatch --test-only scripts/laguna_submit.sh` to validate the script parses without errors. [2 hrs]

□ **Task:** Install MLX and `mlx-lm` locally on the M5: `pip install mlx mlx-lm`. Download `mistralai/Mistral-7B-v0.3` weights in MLX format: `mlx_lm.convert --hf-path mistralai/Mistral-7B-v0.3 --mlx-path models/mistral-7b-mlx`. Confirm the model loads and generates a test completion. This download is ~14GB — start it immediately and work on other tasks while it runs. [1 hrs]

□ **Task:** Run a 10-step smoke fine-tune locally with MLX: `mlx_lm.lora --model models/mistral-7b-mlx --train --data data/finetune/train.jsonl --batch-size 2 --lora-layers 4 --iters 10`. Confirm the run completes, loss decreases over 10 steps, and an adapter file is saved. Log the output to `docs/mlx_smoke_run.txt`. [1 hrs]

□ **Task:** Implement `electoral/llm/trainer.py`: `TrainConfig` dataclass with fields: `base_model`, `lora_rank`, `lora_alpha`, `learning_rate`, `epochs`, `batch_size`, `train_path`, `eval_path`, `output_dir`, `backend` (“`mlx`” or “`hpc`”). Write `load_config(path)` and `save_config(config, path)`. Create `configs/train_r16.json` and `configs/train_r32.json` with the hyperparameters from the development plan. [1 hrs]

Day 2 (Tue — Prompt Engineering & First Training Run)

□ **Task:** Design the prompt template. Write three candidate templates in `docs/prompt_candidates.md` and test each on 5 training examples manually using the base (un-fine-tuned) Mistral 7B locally. Evaluate: does the model output valid JSON? Does the structure make the task obvious? Select the best template and document the reasoning. [2 hrs]

□ **Task:** Implement the selected prompt template in `trainer.py`: `format_prompt(example)` that takes a JSONL record and returns a formatted string. The prompt must include: (i) system context paragraph establishing the political science framing and the selected party; (ii) the event description; (iii) RoBERTa news scores as a labeled JSON block (per-bloc); (iv) X bio-weighted social scores as a labeled JSON block (per-bloc, sourced

from `social_roberta_scores`); (v) the historical analog and shock category; (vi) a closing instruction: “You are analyzing the impact on the `[party]` candidate’s coalition. Output only a JSON object with three keys: `delta_bins_race` (keys: `african_american`, `latino`, `asian`, `white`, `other_race`), `delta_bins_religion` (keys: `evangelical`, `catholic`, `protestant`, `secular`, `jewish`, `muslim`, `other_rel`), and `delta_bins_gender` (keys: `women`, `men`, `other_gender`). Values are one of: `strong_neg`, `mod_neg`, `mild_neg`, `slight_neg`, `neutral`, `slight_pos`, `mild_pos`, `mod_pos`, `strong_pos`. Also output `delta_eff` (scalar): the pipeline will verify it matches $\lambda_1 \sum w_i \Delta \mu_i^{\text{race}} + \lambda_2 \sum v_R \Delta \mu_R^{\text{rel}} + \lambda_3 \sum g_G \Delta \mu_G^{\text{gen}}$. For non-gendered shocks, gender bins may be identical across women/men.” The party name must be substituted from `example["party"]`, never hardcoded. [2 hrs]

□ **Task:** Implement the USC Laguna training path in `trainer.py`: when `backend='hpc'` (parameter name unchanged; targets USC Laguna), use HuggingFace `transformers` + `peft.LoraConfig` + `BitsAndBytesConfig` (4-bit NF4). Implement `train_hpc(config)` that loads the base model in 4-bit, applies LoRA adapters to Q/K/V/O projections, runs `Trainer.train()`, and saves the adapter to `config.output_dir`. [2 hrs]

□ **Task:** Submit the first full USC Laguna training run: `sbatch scripts/laguna_submit.sh` with `configs/train_r16.json`. Monitor the job queue with `squeue -u $USER`. Once the job starts, `tail -f` the SLURM output file to watch the loss curve. The job should complete overnight. If it times out, increase walltime and resubmit. [1 hrs]

□ **Task:** While the USC Laguna job runs: implement `electoral/llm/eval.py`. `EvalResult` dataclass: `shock_id`, `mae_per_bloc` (dict), `mean_mae`, `output_json` (raw model output). `evaluate_model(adapter_path, eval_path)` loads the adapter, iterates through each eval example, runs `estimate_shock`, computes MAE against ground-truth deltas, and returns a list of `EvalResult`. Write `print_eval_summary(results)` that logs a table of MAE per bloc sorted descending. [1 hrs]

Day 3 (Wed — Constrained Decoding & Inference Wrapper)

□ **Task:** Install `outlines`: `pip install outlines`. Confirm it imports correctly alongside the existing HuggingFace stack. Read the `outlines.generate.json` documentation. Write a standalone 10-line proof-of-concept script `scripts/outlines_poc.py` that loads a tiny HuggingFace model (`gpt2`), generates constrained JSON with a simple schema (`{'x': int}`), and prints the result. Confirm the output is always valid JSON regardless of how many times you run it. [1 hrs]

□ **Task:** Convert `ShockResponseData` to a **Pydantic** model in `electoral/artifacts.py` (or a parallel `ShockResponseSchema` Pydantic class if keeping the frozen dataclass for internal use). Define the 10 bloc fields as `Annotated[str, Literal["strong_neg", "mod_neg", "mild_neg", "weak_neg", "neutral", "weak_pos", "mild_pos", "mod_pos", "strong_pos"]]` — one field per canonical bloc. Passing this Pydantic model directly to `outlines.generate.json(model, ShockResponseSchema)` has two critical advantages over

a manual `json.schema()` dict: (i) the LLM output schema and the FastAPI `response_model` are the *same Python object*, so they can never drift out of sync; (ii) Pydantic's `Literal` type maps cleanly to the 9-token enum constraint that constrained decoding enforces. Write a unit test confirming the Pydantic model validates a correct output and rejects an unknown bin token. [1 hrs]

□ **Task:** Implement `electoral/llm/inference.py`. Class `ShockEstimator` with `__init__(self, adapter_path, base_model)` that loads the model once and keeps it in memory. Method `estimate(self, event, intensity)` that: (i) formats the prompt via `format_prompt`; (ii) passes the `ShockResponseSchema` Pydantic model directly to `outlines.generate.json(model, ShockResponseSchema)` — no manual dict construction; (iii) runs generation; (iv) maps bin tokens to numeric deltas via `bin_to_delta`; (v) scales delta values by `intensity`; (vi) constructs and validates `ShockResponseData`; (vii) returns it. The `ShockResponseSchema` Pydantic model is also used as the FastAPI `response_model` in `shock_endpoint.py`, guaranteeing the two schemas are always identical. [3 hrs]

□ **Task:** Write `tests/test_constrained_output.py`: instantiate `ShockEstimator` with the trained adapter. Run `estimator.estimate(event, intensity)` 100 times with 10 different event strings (10 calls each). Assert: (i) every output passes `ShockResponseData.validate()`, (ii) all delta values are finite and in $[-0.15, 0.15]$, (iii) all 10 bloc keys are present, (iv) no call raises an exception. [2 hrs]

□ **Task:** Benchmark latency: time 20 sequential calls to `estimator.estimate()` with a standard Ayatollah event description. Print mean, median, and p95 latency. If p95 exceeds 3 seconds: (i) confirm the model is loaded once (not reloaded per call), (ii) try reducing `max_tokens` in the generator, (iii) try 4-bit quantization if not already active. Document the final p95 in `docs/latency_benchmark.md`. [1 hrs]

Day 4 (Thu — FastAPI Endpoint & Baseline Comparison)

□ **Task:** Implement `electoral/api/shock_endpoint.py`. Use FastAPI lifespan context to load `ShockEstimator` once at startup and store in `app.state.estimator`. Define three routes: `POST /estimate` (body: `EstimateRequest(event: str, intensity: float = 1.0)`, returns `ShockResponseData`), `GET /health` (returns `{"status": "ok", "model": adapter_path}`), `GET /blocs` (returns list of canonical bloc names from `types.py`). Add `CORSMiddleware` allowing `http://localhost:3000` for local development. [2 hrs]

□ **Task:** Add request validation and error handling to the endpoint. If `event` is empty or under 10 characters, return HTTP 422 with message “Event description too short”. If `intensity` is outside $[0.1, 3.0]$, return HTTP 422 with message “Intensity out of range”. Wrap the `estimator.estimate()` call in a try/except: on any unexpected error, return HTTP 500 with a sanitized message (no stack trace in the response body, but log the full trace server-side). [2 hrs]

□ **Task:** Run the three-way baseline comparison on the 2020 eval set. For each method, call the appropriate function on every eval example and collect `EvalResult` lists: (i) news-only: use the `SentimentData` artifact’s news RoBERTa scores mapped through the elasticity regression; (ii) social-only: same but using only platform scores; (iii) unified LLM: use `estimator.estimate()`. Compute mean MAE per bloc and overall. Save to `artifacts/baseline_comparison.json` and `docs/baseline_comparison.md` with a formatted table. [2 hrs]

□ **Task:** Write `tests/test_endpoint_latency.py`: start the FastAPI app as a subprocess with `subprocess.Popen`, wait for the health check to return 200, then send 5 POST requests to `/estimate` with the Ayatollah event description. Assert each response: (i) HTTP 200, (ii) valid JSON, (iii) parses into `ShockResponseData` without error, (iv) response time under 3 seconds. Teardown: terminate the subprocess. [2 hrs]

Day 5 (Fri — Hyperparameter Decision, Kernel Wiring & PR)

□ **Task:** Compare the $r = 16$ and $r = 32$ eval results. Decision rule: adopt $r = 32$ if mean MAE improves by more than 5% across all blocs. If neither run is satisfactory (mean MAE > 0.05 vote-share points), consider the Qwen2.5 14B fallback: re-run the smoke MLX test with `Qwen/Qwen2.5-14B-Instruct` and compare structured output reliability. Document the final model and rank decision in `DECISIONS.md` with the supporting MAE numbers. [2 hrs]

□ **Task:** Retrieve the winning adapter weights from USC Laguna scratch using `scp`, not Synthing. USC Laguna scratch nodes frequently kill background daemons including Synthing; relying on it for model weight transfer risks silent failures or corrupted weights from race conditions. Run: `scp -r laguna:$SCRATCH/electoral-equilibrium/adapters/mistral-7b-electoral/adapters/`. Verify the adapter loads correctly on M5. Tag the version in `adapters/mistral-7b-electoral/VERSION` with the USC Laguna job ID, training config path, and eval MAE. After verifying the transfer, add the adapter directory to Synthing for local machine availability. [1 hrs]

□ **Task:** Wire `electoral/kernels/finetune.py`: implement `build_llm_finetune(config)` that reads `LLMFineTuneData` from the artifact store (or constructs it from the adapter directory if already trained). Wire into `stages.py`. The finetune stage is idempotent: if an adapter already exists at the configured path, skip training and log “adapter found, skipping training”. [2 hrs]

□ **Task:** Wire `electoral/kernels/shock.py`: implement `build_shock_response(config, event, intensity)` that calls `estimate.shock`, `bootstrap_cov`, and `solve_rebalanced` in sequence, wraps results in `ShockResponseData` and `EquilibriumData` payloads, and writes both artifacts. Run the full pipeline smoke config end-to-end including the shock stage. [2 hrs]

□ **Task:** Weekly PR: `feature/week5-llm-finetune`. PR description must include: (i) the

three-way MAE comparison table, (ii) the model and rank decision with justification, (iii) USC Laguna job IDs and adapter path, (iv) a sample of 3 eval examples showing input event, predicted deltas, and ground-truth deltas side by side. [1 hrs]

□ **Task:** Brief supervision check-in with Prof. Espinosa: share 3–4 eval examples where the model’s predictions diverge most from ground truth. Get domain perspective on whether divergences reflect real political dynamics. Juhi decides on any model adjustments. [2 hrs]

6 Week 5 — Stochastic Optimization & Monte Carlo (40 hrs)

Goal. By Friday, the CVXPY optimizer rebalances voter distributions after a shock, 10,000+ Monte Carlo simulations run on USC Laguna, and the `SimulationData` artifact is produced and validated.

End condition.

- ☐ Optimizer produces feasible weights for all test shocks.
- ☐ Monte Carlo win probability converges within ± 0.005 between $N = 5,000$ and $N = 10,000$.
- ☐ All optimization and simulation tests pass.
- ☐ PR merged.

Day 1 (Mon — Schema Migration, Post-Shock Optimization)

☐ **Task:** Add `win_probability_low` and `win_probability_high` to `SimulationData` before writing any Monte Carlo code. The current `SimulationData` only stores `win_probability` (a point estimate) and `percentiles` (per-bloc coalition-weight distributions). It is missing the 90% confidence interval on the *win-probability* itself, which is a named deliverable in `DECISIONS.md` and required by the WinGauge UI logic:

- `win_probability_low`: float — 5th percentile of $\Pr(\tilde{\mu}_{\text{eff}}(\mathbf{w}^{(n)}) \geq V_{\text{eq}})$ across the N ILR draws.
- `win_probability_high`: float — 95th percentile.

These come from propagating Σ_{Δ} uncertainty through the ILR back-transform (not Bernoulli variance). The WinGauge fires “*No feasible path*” when `win_probability_high` < V_{eq} and “*Uncertain path*” when `win_probability_low` < $V_{\text{eq}} \leq \text{win_probability_high}$.

Add both fields to the frozen dataclass, update `validate()` to check $0 \leq \text{low} \leq \text{high} \leq 1$, update `from_dict` / `to_dict`, update the `run_simulations` stub in `stages.py` to provide placeholder values of 0.0 and 1.0 respectively, and update `test_artifact_roundtrip.py`. Do *not* start the Monte Carlo kernel until this schema change is committed and all tests pass. [1 hrs]

☐ **Task:** Implement `electoral/optimization/simplex.py: project_simplex(v)` using the Duchi et al. (2008) $O(n \log n)$ algorithm. The algorithm: sort v descending, find the largest ρ such that $v_{(\rho)} - \frac{1}{\rho}(\sum_{j \leq \rho} v_{(j)} - 1) > 0$, set $\theta = \frac{1}{\rho}(\sum_{j \leq \rho} v_{(j)} - 1)$, return $\max(v - \theta, 0)$. Write three unit tests: (i) already-valid simplex vector returns unchanged, (ii) negative components are zeroed and remainder redistributed, (iii) output always sums to 1.0 within `tol=1e-9`. [3 hrs]

☐ **Task:** Implement `electoral/optimization/cvx.py: solve_rebalanced(mu_tilde, cov_delta, target)` using CVXPY with the **max P(win) objective** (not min-variance). Formulate the Sharpe-ratio objective: `numerator = mu_tilde @ w - target` (linear in

w); denominator = `cp.sqrt(cp.quad_form(w, cov_delta))` (convex in w); objective = `cp.Maximize(cp.ratio(numerator, denominator))`. Declare `problem.solve(qcp=True)` (Disciplined Quasiconvex Programming). **Assert** `problem.is_dcp(qcp=True) == True` **before solving** — if this assertion fails, the objective is not quasi-convex and the solver cannot guarantee the global optimum. This assertion must be a unit test. On `SolverError` or infeasibility, return `EquilibriumData` with `feasible=False`. [4 hrs]

□ **Task:** Implement `solve_equal_weight_rebalanced(blocs, delta_w)` as a fallback in `cvx.py`: if the QP is infeasible after 5 retries, return an equal-weight distribution with `feasible=False` and `target_met=False`. Log a warning. This ensures the web app never crashes due to an infeasible shock input. [2 hrs]

Day 2 (Tue — Covariance Bootstrap & Full Shock Chain)

□ **Task:** Implement `electoral/models/bootstrap.py` using **Ledoit-Wolf shrinkage** rather than raw bootstrap covariance. With only 6–8 election cycles, bootstrapping a 10×10 covariance matrix yields a rank-deficient result (≤ 7 degrees of freedom for 55 parameters). Applying `psd_repair` to a rank-deficient matrix just adds noise — the optimizer would be minimizing against your artificial ϵI injection rather than historical electoral reality.

Use `sklearn.covariance.LedoitWolf` instead: it applies an analytically optimal shrinkage toward a diagonal target, is specifically designed for the $p > n$ regime, and produces a well-conditioned, full-rank covariance estimate. Usage: instantiate `LedoitWolf(assume_centered=False)`, call `.fit(delta_matrix)` where `delta_matrix` has shape `(n_cycles, 10)`, read `.covariance_` — always full-rank, no `psd_repair` needed. Keep `bootstrap_cov` as an alternative for comparison; document in `DECISIONS.md` that Ledoit-Wolf is the primary estimator. Write unit tests: (i) output is symmetric and PSD, (ii) condition number is finite and below 10^6 , (iii) same input produces identical output. [3 hrs]

□ **Task:** Implement the sensitivity-weighted covariance variant `bootstrap_cov_weighted(panel_df, social_elasticities, n_bootstrap=1000, seed=42)`: weight each cycle's contribution by its mean absolute social media elasticity across platforms, so cycles with stronger social media reactions contribute more to the covariance estimate. This fuses the social media signal directly into the stochastic model. Store both variants; the paper will compare them. [2 hrs]

□ **Task:** Integrate the full shock response chain in `electoral/kernels/shock.py`: `build_shock_response(config, event_description, intensity)`. Step 1: call `estimate_shock` to get `delta_mu` dict ($\Delta\mu_i$ per bloc). Step 2: load μ and $w^{(0)}$ from the `BaselinePortfolioData` artifact. Step 3: compute $\tilde{\mu} = \mu + \Delta\mu$. Step 4: call `bootstrap_cov` to get Σ_Δ . Step 5: call `solve_rebalanced(mu_tilde, cov_delta, target)` to produce `EquilibriumData` (which stores both $\tilde{w}$ and $\tilde{\mu}$). Step 6: construct and validate `ShockResponseData` wrapping `delta_mu` and the covariance. Wire into `stages.py`. [3 hrs]

Day 3 (Wed — Monte Carlo Engine)

□ **Task:** Implement `electoral/simulation/montecarlo.py: run_simulations(w_star, mu_tilde, cov_delta, target, n, seed)`. Use the **Logistic-Normal with ILR parameterization** — not Dirichlet (which forces negative off-diagonal covariances) and not Gaussian-then-project (which destroys covariance structure asymmetrically at simplex boundaries).

Implementation steps:

- Construct a Helmert contrast matrix $V \in \mathbb{R}^{(K-1) \times K}$.
- Map $\mathbf{w}^*$ to ILR coordinates: $\mathbf{z}^* = V^\top \log(\mathbf{w}^*)$
- Propagate Σ_Δ to ILR space via the Jacobian J of the ILR transformation: $\Sigma_{\text{ILR}} = J \Sigma_\Delta J^\top$
- Draw N samples: $\mathbf{y}^{(n)} \sim \mathcal{N}(\mathbf{z}^*, \Sigma_{\text{ILR}})$
- Back-transform via softmax: $w_i^{(n)} = \exp(y_i^{(n)}) / \sum_j \exp(y_j^{(n)})$
- Compute win flags and confidence interval bounds (5th and 95th percentiles)

If $w_i^* = 0$ exactly for any bloc, report it as `infeasible_bloc` in `EquilibriumData` and exclude it from the simulation — do *not* floor to 0.01 (flooring distorts log-ratio space non-linearly).

Required test: `test_montecarlo_degenerate` — with $w_0^* = 0.99$, $w_{1..4}^* = 0.0025$, assert all N samples sum to 1.0 within `tol=10-9` and `win_probability` is in $[0, 1]$. [5 hrs]

□ **Task:** Implement `project_simplex_batch(W)` in `simplex.py`: takes an $(N \times d)$ matrix and returns each row projected onto the probability simplex. This function is a **fallback utility** used only when the CVXPY optimizer returns an infeasible solution and equal-weight distribution is substituted — it is *not* the primary Monte Carlo sampling path (which uses Dirichlet and produces simplex-valid samples directly).

Implement Duchi's $O(n \log n)$ algorithm per row. A Python `for` loop over rows is acceptable and correct here: at $N = 10,000$, $d = 10$, the dominant cost is the sort, and Python overhead per row is negligible. Vectorizing the dynamic per-row threshold ρ in pure NumPy is brittle without a loop — do not attempt it. If the fallback path ever becomes a throughput bottleneck (it won't during the SRP), use `numba.jit` to compile the inner loop.

Tests: (i) all rows sum to 1.0 within `tol=1e-9`, (ii) all values non-negative, (iii) already-valid row returns unchanged, (iv) $N = 10,000$ completes in under 5 seconds on M5 (loop is fine). [2 hrs]

□ **Task:** Write `scripts/run_montecarlo_laguna.sh`: SLURM batch script for USC Laguna requesting 8 CPU cores, 16GB RAM, 2-hour walltime. The script loads a shock event from `artifacts/shock_{id}.json`, runs `python -m electoral.simulation.montecarlo` with $N = 50,000$, and saves `SimulationData` to `artifacts/sim_{id}.json`. Submit for the Ayatollah event as a test job. Confirm the job completes and produces a valid artifact. [1 hrs]

□ **Task:** Write `tests/test_montecarlo.py`: (i) smoke run $N = 100$ completes in under 2 seconds on M5, (ii) win probability is in $[0, 1]$, (iii) with μ all equal to 0.8 and `target=0.535`, win probability > 0.95 , (iv) with μ all equal to 0.2, win probability < 0.05 , (v) two runs with same seed produce identical `SimulationData.to_dict()` outputs, (vi) percentiles list has exactly 5 values per bloc in $[0, 1]$. [1 hrs]

Day 4 (Thu — Convergence Tests, Sensitivity Analysis & Kernel Wiring)

□ **Task:** Write `tests/test_mc_convergence.py`: run simulations at $N = 1,000$, $N = 5,000$, and $N = 10,000$ with the same seed. Assert the $N = 5,000$ and $N = 10,000$ win probability estimates differ by less than 0.005. Also assert the $N = 1,000$ estimate is within 0.02 of the $N = 10,000$ estimate (looser bound). Log the three estimates side by side. [2 hrs]

□ **Task:** Write `tests/test_optimization.py`: (i) a feasible shock (small delta, weak covariance) returns weights summing to 1.0 with `feasible=True` and `target_met=True`, (ii) an extreme shock that pushes the portfolio below target returns `target_met=False` and logs the relaxation steps, (iii) `project_simplex` output always non-negative and sums to 1.0, (iv) same inputs produce identical `EquilibriumData.to_dict()` across two calls. [2 hrs]

□ **Task:** Implement a shock sensitivity analysis script `scripts/sensitivity_analysis.py`: for a given shock event, vary the intensity parameter from 0.5 to 2.0 in steps of 0.25 and plot how win probability and per-bloc weights change. Save the plot as `artifacts/sensitivity- $\{shock\_id\}$ .png`. Run on the Ayatollah event as a case study. This figure will be useful in the paper. [2 hrs]

□ **Task:** Wire `electoral/kernels/optimize.py` into `stages.py`: `build_optimization(config, shock_response_artifact)` that loads the `ShockResponseData`, runs `solve_rebalanced`, and writes `EquilibriumData`. Run the full pipeline smoke config end-to-end and confirm all seven artifact files exist and validate. [2 hrs]

Day 5 (Fri — Metrics, Reporting & PR)

□ **Task:** Implement `electoral/metrics/performance.py`. Required functions: `win_probability(sim_data)` (reads from `SimulationData`), `equilibrium_gap(weights, mu, target)` returning $\mathbf{w}^\top \mu - V_{\text{eq}}$ (positive = above threshold), `bloc_delta_summary(baseline_weights, rebalanced_weights)` returning a dict of signed deltas per bloc sorted by absolute magnitude. All functions must accept and return JSON-native types. [2 hrs]

□ **Task:** Implement `electoral/metrics/tables.py`: write `build_shock_summary_table` (takes `shock_ids` and `config`) that iterates over shock event IDs, loads the corresponding `EquilibriumData` and `SimulationData` artifacts, and assembles a summary table with

columns: shock id, category, win probability, equilibrium gap, top-moving bloc, and top-moving delta. Return as a JSON-serializable list of dicts. [2 hrs]

□ **Task:** Implement `electoral/reporting/export.py`: `export_json(path, data)`, `export_csv(path, rows, fieldnames)`, and `export_latex_table(path, rows, caption, label)` that produces a booktabs-style LaTeX table fragment. Run the summary table for all 25+ shocks and save to `artifacts/shock_summary_table.csv` and `artifacts/shock_summary_table.tex`. [2 hrs]

□ **Task:** Weekly PR: `feature/week6-optimization-montecarlo`. Include the convergence test output, the sensitivity analysis plot for the Ayatollah event, and the shock summary table (first 10 rows) in the PR description. [1 hrs]

□ **Task:** Brief supervision check-in with Prof. Espinosa: share win probability estimates for 4–5 historical shocks. Get domain input on whether results align with known electoral outcomes. Juhi decides which covariance variant to use in the paper and documents the decision. [1 hrs]

Day 4.5 — Synthetic Data Augmentation Pipeline

Task: Design a four-axis event taxonomy for synthetic fine-tuning data generation. Axes: (i) domain (econ, foreign, social, candidate, institutional, scandal, exogenous), (ii) ideological valence (far_left through far_right, libertarian, populist, bipartisan), (iii) electoral effect (helps_dem, helps_rep, splits, realigns, neutral), (iv) party perspective (democrat, republican). Key design principle: ideological valence does NOT determine electoral effect — a far-left event can hurt Democrats with moderates. Store as `configs/synthetic_events.json` (62 balanced seeds) and `docs/synthetic_event_taxonomy.md`. Balance targets: helps_dem/helps_rep both $\geq 15\%$, both party perspectives $\geq 40\%$. [2 hrs]

Task: Implement `scripts/synthetic/generate_deepseek.py`: reads seed events from `configs/synthetic_events.json`, expands each seed into N training records via the DeepSeek API (OpenAI-compatible, `deepseek-chat` model), validates every record against the 9-token bin schema and canonical bloc keys, and writes to `data/finetune/candidates.jsonl`. Key implementation details: batch size of 5 records per API call (20 causes truncation at 4k tokens), `max_tokens=8000`, retry logic with exponential backoff. Generated 1,237 records from 62 seeds with only 3 rejections. Effect balance: splits 48%, helps_dem 23%, helps_rep 16%, neutral 8%, realigns 5%. [2 hrs]

Task: Implement `scripts/synthetic/gemini_review.py`: three-stage review pipeline using Gemini 2.5 Flash as a political plausibility reviewer. For each candidate record, Gemini checks: (i) direction sanity per bloc, (ii) delta_eff sign coherence, (iii) cross-bloc logic (evangelical vs secular), (iv) magnitude realism. Triages into APPROVE / REVISE (with corrected labels) / HUMAN_REVIEW. Revisions are audited in `data/finetune/revisions.jsonl`. A 5%

random spot-check of approved records is routed to the human queue regardless of verdict. Human queue (`data/finetune/human_review.queue.jsonl`) is adjudicated in a separate Opus/human pass during low-usage hours. Results: 579 approved, 647 queued for human review (Gemini 2.5 Flash is conservative by design). [2 hrs]

Task: Merge Gemini-approved synthetic records into the training set. Strip `_review` and `_seed_meta` metadata fields before merging. Rebuild 80/20 train/eval split with seed=42. Final dataset: 1,136 synthetic records (577 original + 559 Gemini-approved). Retrain Mistral 7B r=16 adapter on the expanded dataset. Document in `DECISIONS.md`: synthetic data pipeline rationale, DeepSeek generation settings, Gemini review thresholds, human review queue process, and the key insight that ideological valence and electoral effect are independent axes. [2 hrs]

Task: NEP-weighted baseline coalition formula. Discovered that the original `build_baseline_portfolio` used CVXPY to derive coalition weights $\mathbf{w}^{(0)}$, producing degenerate results (asian=0, latino=0, african_american=0.51). Replaced with the principled formula: $w_i = (\text{NEP}_i \times \mu_i) / \sum_j (\text{NEP}_j \times \mu_j)$, where NEP shares are from Edison Research 2024 National Exit Poll (actual voter shares, not census). Values: white=0.63, african_american=0.12, latino=0.15, asian=0.05, other_race=0.05. Resulting $\mathbf{w}^{(0)}$: white=0.528, african_american=0.189, latino=0.180, asian=0.051, other_race=0.052. Baseline μ_{eff} dropped from 0.716 (equal weights) to 0.605 (NEP-weighted). Stored in `configs/base.json` as `bloc_population_shares`. [2 hrs]

7 Week 6 — Web Application (40 hrs)

Goal. By Friday, the core public web application is live end-to-end with SSE streaming and deployed on Modal + Vercel. The analyst dashboard moves to Week 8 Days 1–2 to give it the time it actually requires.

End condition.

- ☐ Public app renders per-bloc shifts progressively via SSE.
- ☐ Audit log records every `/estimate` call with latency.
- ☐ Modal + Vercel deployment live with public URL.
- ☐ PR merged.

Day 1 (Mon — Next.js Setup & Typed API Client)

☐ **Task:** Initialize the Next.js app: `npx create-next-app webapp --typescript --tailwind --app --no-src-dir`. Remove all boilerplate page content. Set up `webapp/.env.local`: `NEXT_PUBLIC_API_URL= http://localhost:8000`. Confirm `npm run dev` starts without errors and the browser shows a blank page. [1 hrs]

☐ **Task:** Install frontend dependencies: `npm install recharts zod @radix-ui/react-slider @radix-ui/react-collapsible lucide-react`. These cover charting, schema validation, the intensity slider, the “How it works” expandable, and icons respectively. [1 hrs]

☐ **Task:** Define TypeScript types in `webapp/lib/types.ts`: all existing payload types (`ShockResponseData`, `EquilibriumData`, `SimulationData`, `EstimateResponse`) plus new dashboard types: `AuditEntry` (`id`, `timestamp`, `event_text`, `party`, `intensity`, `win_prob`, `feasible`, `llm_ms`, `optimizer_ms`, `montecarlo_ms`), `CoverageCell` (`bloc_id`, `cycle`, `quality`, `vote_share`, `sources`), and `TrainingRun` (`run_id`, `epochs`, `train_loss`, `val_loss`, `mae`). Add `Party = "democrat" | "republican"` as a shared literal type. All types must mirror their FastAPI response models exactly. [1 hrs]

☐ **Task:** Implement Zod schemas in `webapp/lib/schemas.ts` for each type. These are used to validate API responses at runtime so the frontend fails loudly rather than rendering NaN or undefined. Export `EstimateResponseSchema.parse(data)` as the single validation entry point. [1 hrs]

☐ **Task:** Implement `webapp/lib/api.ts`: `estimateShock(event: string, intensity: number): Promise<EstimateResponse>` that POSTs to `/estimate`, parses the response with the Zod schema, and throws a typed `ApiError` (with `message` and `status`) on any failure. Implement `getBlocs(): Promise<string[]>` and `healthCheck(): Promise<boolean>`. [2 hrs]

□ **Task:** Write a smoke integration test for the API client: in `webapp/lib/__tests__/api.test.ts`, mock `fetch` using `jest.fn()`, provide a valid `EstimateResponse` fixture, and assert: (i) the correct URL is called, (ii) the response parses without error, (iii) a malformed response throws `ApiError`. [2 hrs]

Day 2 (Tue — ShockInput Component & State Management)

□ **Task:** Implement the page-level state in `webapp/app/page.tsx` using `useState`: `party` ("democrat" | "republican", default "democrat"), `event` (string), `intensity` (number, default 1.0), `loading` (boolean), `error` (string — null), `deltaBins` (`Record<string, string>` | null), `shifted` (`Record<string, number>` | null), `equilibrium` (`EquilibriumData` | null), `simulation` (`SimulationData` | null).

Implement `handleSubmit`: open an SSE `EventSource` to `/estimate/stream`; on `event: deltas` set `deltaBins` and `shifted` (populates `ShockNarrative` immediately, and renders `CoalitionChart` with opaque bars only — rebalance layer is absent until the next event); on `event: equilibrium` set `equilibrium` (adds the translucent rebalance layer to `CoalitionChart` in-place — no re-render of the opaque bars); on `event: simulation` set `simulation` (populates `WinGauge`); on `[DONE]` set `loading=false`.

Layout: `ShockNarrative` above the chart, full width. `CoalitionChart` below it, full width (the chart is self-contained with both layers). `WinGauge` to the right on desktop, below on mobile. The progressive reveal — opaque bars first, translucent overlay second, gauge last — gives the user something to read at each SSE step. [3 hrs]

□ **Task:** Implement `webapp/components/ShockInput.tsx`. Props: `party`, `setParty`, `event`, `setEvent`, `intensity`, `setIntensity`, `onSubmit`, `loading`. Render: (i) a two-button toggle labeled **Democrat** / **Republican**, styled blue/red respectively, that sets `party` state — this is the first control the user sees; (ii) a `<textarea>` with placeholder “Describe a hypothetical political event...”, minimum 3 rows, full width; (iii) a labeled `Slider` (Radix UI) for intensity 0.5–2.0 in steps of 0.1; (iv) a submit `<button>` showing a spinner when loading. Style with Tailwind. The party selection must be prominent — it determines the entire analysis direction. [3 hrs]

□ **Task:** Add five preset shock buttons below the textarea. Each button pre-fills the event field with a template string. Presets: (i) Security: “An assassination attempt is made on the Democratic candidate”; (ii) Geopolitical: “The US enters a major armed conflict in the Middle East”; (iii) Moral/Scandal: “A major financial scandal involving the Republican candidate is revealed”; (iv) Electoral Surprise: “A major October Surprise leak dominates the news cycle”; (v) Economic: “A sudden recession is declared in the final month of the campaign”. Style each button with its shock category color. [2 hrs]

□ **Task:** Implement the error display: below the submit button, show a styled red alert box when `error` is non-null. The message should be user-friendly (“The estimation service is

unavailable. Please try again.”) rather than exposing the raw API error. Log the raw error to the browser console for debugging. [1 hrs]

Day 3 (Wed — Prediction Panel: ShockNarrative & CoalitionChart)

□ **Task:** Implement `webapp/components/ShockNarrative.tsx`. Props: `deltaBins: Record<string, string>`, `party: Party`. This component renders the plain-language **prediction** — what the model expects will happen to each voter group after the shock, in words the user can read without knowing what a delta bin is.

Logic: map each `delta_bin` token to a phrase pair (direction + magnitude): `strong_neg` → “significant loss among”, `mild_pos` → “modest gain with”, `neutral` → “no meaningful change among”, etc. Group blocs by direction (gaining vs. losing) and render two sentences:

- “This shock is predicted to **hurt** the [party] coalition’s standing with: [Evangelicals (significant), Catholics (mild)].”
- “And **help** with: [Latino voters (moderate), Women (mild)].”

If all blocs are neutral, render: “The model predicts minimal coalition impact from this event.” The narrative populates immediately when the `event: deltas` SSE event fires — the user sees the prediction in words before the charts finish rendering. [3 hrs]

Task: Implement `webapp/components/CoalitionChart.tsx` — the single unified chart that shows both the prediction and the rebalance in one view. Props: `baseline: Record<string, number>` (pre-shock loyalty μ_i), `shifted: Record<string, number>` (post-shock loyalty $\tilde{\mu}_i$ — the **prediction**), `rebalanced: Record<string, number> | null` (optimizer coalition weights $\tilde{w}_i$ — the **rebalance**; null until the `event: equilibrium` SSE event fires), `feasible: boolean`, `party: Party`.

Visual design — three series on one horizontal bar chart:

- **Opaque bars (prediction):** post-shock loyalty $\tilde{\mu}_i$, rendered at full opacity. Color: blue for Democrat, red for Republican. This is what the model says *will happen* to each group’s loyalty after the shock.
- **Translucent bars (rebalance):** optimizer coalition weight $\tilde{w}_i$, rendered at 35% opacity (`fillOpacity=0.35`) on top of the opaque bars using the same color. This is what the model says the coalition *should look like* to still win. Absent until `event: equilibrium` fires; the opaque bars are visible and readable while the rebalance is loading.
- **Reference tick marks:** a vertical `ReferenceLine` at μ_i (pre-shock baseline) per bloc, rendered as a thin gray tick, so users can see how far the shock moved loyalty from the starting point.

Why same axis works: both $\tilde{\mu}_i$ and $\tilde{w}_i$ are in $[0, 1]$. $\tilde{\mu}_i$ is within-bloc loyalty (typically 0.15–0.90). $\tilde{w}_i$ is coalition weight (typically 0.03–0.30). They occupy different ranges and are visually distinct. A clear legend is required: “— Prediction (loyalty shift)” and “— Rebalance (coalition weight, translucent)”. The tooltip on hover must label them explicitly with different units.

Sorting: sort blocs by `Math.abs(shifted[bloc] - baseline[bloc])` descending so the most-moved blocs appear at the top.

Labels: on each opaque bar, show the signed loyalty delta (+8pp, -3pp) in green/red. On each translucent bar, show the coalition weight as a percentage (14%).

Feasibility: if `feasible=false` and `rebalanced` is non-null, replace the translucent bars with a striped red pattern and add a banner: “No feasible coalition path under this shock.”
[5 hrs]

□ **Task:** Add a section header above `CoalitionChart` with two sub-labels that map to the two visual layers: “**What will happen**” (opaque, bold) and “**Most likely rebalance**” (translucent, muted) — styled as a color-coded legend inline with the header so users understand what each layer means before reading the chart. Below the chart, add a summary line: “Equilibrium status: **MET / NOT MET**” in green/red, plus the gap formatted as “+3.2 pp above target”.
[1 hrs]

Day 4 (Thu — WinGauge, SSE Streaming & Audit Log)

□ **Task:** Implement `webapp/components/WinGauge.tsx`. Props: `winProbability: number`, `percentiles: Record<string, number[]>`. Render a semicircular SVG gauge (viewBox 0 0 200 110). Draw the arc as three colored segments: red (0–45%), yellow (45–55%), green (55–100%). Draw a needle rotating from 0° (left) to 180° (right) based on `winProbability`. Animate the needle with a CSS transition on value change (`transition: transform 0.8s ease-out`).
[2 hrs]

□ **Task:** Add a percentile distribution strip below the gauge: a compact horizontal box-plot showing p5, p25, p50, p75, p95 per bloc from `SimulationData.percentiles`. Use a small `Recharts ComposedChart` with a `Bar` for the IQR and `ErrorBar` for p5–p95 whiskers. [1 hrs]

□ **Task:** Add the **market prior display** to the `WinGauge` component. After the model win probability renders, call `GET /api/market-prior?party=&event=` (new FastAPI route that queries Polymarket and Manifold for the nearest relevant contract via `electoral/markets/aggregator.py`). Display the volume-weighted result as “Market consensus: **X%**” in a muted gray style directly below the model estimate. If no active contract is found, show “No active market” in italics. The juxtaposition is the key epistemic display: the market aggregates all current public information; the model shows how the coalition structure shifts under the hypothetical. Add disclaimer: “Model: $N = 10,000$ simulations. Market: live Polymarket/Manifold prices.”
[2 hrs]

□ **Task:** Implement **SSE streaming** on the FastAPI backend. Add `GET /estimate/stream` using `StreamingResponse` with `media_type="text/event-stream"`. The generator yields three events: `event: deltas` (LLM output), `event: equilibrium` (optimizer), `event: simulation` (Monte Carlo). Wrap the optimizer and Monte Carlo calls correctly for their respective concurrency models:

- **CVXPY optimizer** — use `ProcessPoolExecutor`: `loop.run_in_executor(process_pool, solve_rebalanced, ...)`. CVXPY’s underlying C solvers (SCS/ECOS/OSQP) are not thread-safe and will segfault under `ThreadPoolExecutor`. Passing `None` defaults to a thread pool — never pass `None`. Instantiate `ProcessPoolExecutor(max_workers=1)` once at startup and reuse it.
- **Monte Carlo** (pure NumPy) — use `ThreadPoolExecutor`: `loop.run_in_executor(thread_pool, run_simulations, ...)`. NumPy releases the GIL during array operations; a thread pool is safe and lower-overhead than spawning a new process.

Each stage starts as soon as the previous one emits its SSE event. Expected latency improvement: ~30% on total round-trip. Update `webapp/lib/api.ts`: add `estimateShockStream(event, intensity, onDeltas, onEquilibrium, onSimulation)` that opens an `EventSource`, calls each callback as its event fires, and closes on `[DONE]`. [3 hrs]

□ **Task:** Implement **audit logging**. Create `electoral/api/audit.py`: `AuditLogger` class connecting to `data/audit.duckdb`. On first connect, run: `CREATE TABLE IF NOT EXISTS estimates (id INTEGER PRIMARY KEY, timestamp TIMESTAMP, event_text VARCHAR, intensity FLOAT, deltas_json VARCHAR, feasible BOOLEAN, target_met BOOLEAN, win_prob FLOAT, llm_ms INTEGER, optimizer_ms INTEGER, montecarlo_ms INTEGER, backend VARCHAR)`. Implement `log_estimate(...)` inserting one row per call. Wire into both `/estimate` and `/estimate/stream` endpoints. Add `GET /api/audit?limit=100` returning the most recent rows as JSON. Verify: call the endpoint 5 times, query DuckDB directly, confirm 5 rows exist. [2 hrs]

Day 5 (Fri — Polish, Deployment & PR)

□ **Task:** Assemble `webapp/app/page.tsx`: two-column layout on desktop, single-column on mobile. Add a sticky header with project name, a one-sentence description, and a link to the methodology document. Add the “How it works” expandable (`@radix-ui/react-collapsible`) and the research disclaimer. Test on iPhone viewport (375px). [2 hrs]

□ **Task:** Add a “Share this result” button that serializes `party`, `event`, and `intensity` as URL query params and copies to clipboard. Test that the URL pre-fills the full form including party selection on reload. [1 hrs]

□ **Task:** Deploy: `modal deploy deploy/modal_app.py`. Connect Vercel to the GitHub repo, set `NEXT_PUBLIC_API_URL` to the Modal URL. Confirm the full public end-to-end flow works for both party modes. Scaffold `deploy/hpc_vllm.sh` and `deploy/backend_router.py` (leave deactivated). [2 hrs]

□ **Task:** Weekly PR: `feature/week7-webapp`. Include: three public app screenshots (empty

state, Democrat Ayatollah result, Republican result, mobile layout) and the live Modal + Vercel URLs. Tag “demo-ready”. **Analyst dashboard is Week 8 Days 1–2; do not attempt it here.** [1 hrs]

☐ **Task:** Supervision check-in with Prof. Espinosa: live demo via public URL. Walk through the Ayatollah shock in both party modes. Collect domain feedback on result framing and disclaimer wording for use in Week 8 paper writing. [2 hrs]

8 Week 7 — Analyst Dashboard, Hardening & Deliverables (40 hrs)

Goal. By Friday, the analyst dashboard is live, all integration tests pass, the pipeline is replication-ready, and the paper and poster are complete. The dashboard was moved here from Week 7 to give it realistic time: SSE wiring, DuckDB, password auth, and six data-backed panels is a two-day build, not one.

End condition.

- ☐ Analyst dashboard live at `/dashboard` with all six panels.
- ☐ All six integration tests pass on a clean checkout.
- ☐ Replication package complete and documented.
- ☐ `CONTINUOUS_UPDATE.md` written; `just continuous` runs clean.
- ☐ Research paper draft submitted to SRP committee — must include: (i) formal quasiconvexity proof for the DQCP objective; (ii) explicit additive independence assumption statement; (iii) additive vs. raked model comparison on held-out cycles; (iv) MMD + PCA + PCD synthetic data validation report.
- ☐ Poster submitted to SRP.
- ☐ Final PR merged and tagged `v1.0`.

Day 1 (Mon — Analyst Dashboard: Auth, Backend Routes & Panels 1–3)

☐ **Task:** Implement dashboard authentication. Create `webapp/app/dashboard/page.tsx`: render a login form when no valid `dashboard_session` cookie exists; on submit POST to `/api/dashboard/auth` which compares against the `DASHBOARD_PASSWORD` env var and sets an 8-hour session cookie. Implement `webapp/app/api/dashboard/auth/route.ts` as the Next.js route handler. On success, render the dashboard shell with a `<DashboardNav>` sidebar (links to each panel, logout button). [2 hrs]

☐ **Task:** Implement the three FastAPI backend routes for dashboard data and wire them in `shock_endpoint.py`. GET `/api/coverage`: load the `VoterPanelData` artifact and return a JSON matrix of per-cell quality ("`multi-source`", "`single-source`", "`imputed`", "`missing`") with raw `vote_share` and sources. GET `/api/sentiment-dist?category=`: load `SentimentData` and return per-bloc elasticity score arrays, optionally filtered by shock category. GET `/api/bio-coverage`: return per-shock bio classifier coverage counts (keyword, `SetFit`, fallback). [2 hrs]

☐ **Task:** Implement **Panel 1 — Data Coverage Matrix** in `CoverageMatrix.tsx`: a `Recharts ScatterChart` rendered as a heatmap (bloc × cycle), colored dark blue (multi-source) through light blue (single) through gray (imputed) to white (missing). Tooltip shows raw `vote_share` and sources. [2 hrs]

□ **Task:** Implement **Panel 2 — RoBERTa Score Distributions:** a `ComposedChart` showing min, p25, median, p75, max per bloc across all shocks; shock category dropdown re-fetches with `?category=`. Implement **Panel 3 — Bio Classifier Coverage:** stacked `BarChart`, one column per shock, three segments (keyword / SetFit / fallback). Both panels share a two-column grid row. [2 hrs]

Day 2 (Tue — Analyst Dashboard: Backend Routes & Panels 4-6)

□ **Task:** Implement the three remaining FastAPI backend routes. `GET /api/training-logs:` read HPC training log files from `rawdata/hpc_logs/` (synced via Syncthing from HPC) and return per-epoch train/val loss arrays per run ID. `GET /api/convergence:` load the most recent `SimulationData` artifact and return win probability at $N = 1,000, 5,000, 10,000$ with p5/p95 bands. `GET /api/audit?limit=100&search=:` query `data/audit.duckdb` with an optional `WHERE event_text LIKE %search%` clause; return rows with all columns including party. [2 hrs]

□ **Task:** Implement **Panel 4 — Fine-Tuning Loss Curves:** Recharts `LineChart` with train (solid) and val (dashed) series, horizontal reference line at the 2020 MAE, run-selector dropdown. Implement **Panel 5 — Monte Carlo Convergence:** `AreaChart` with p5/p95 shaded band, dashed final estimate line, ± 0.005 tolerance band, Pass/Fail badge. [2 hrs]

□ **Task:** Implement **Panel 6 — Audit Log Query Interface:** sortable table with columns timestamp, event (40-char truncated), party, intensity, win probability, feasibility, total latency. Free-text search box. Row-expand shows the full per-bloc delta vector as a mini `CoalitionChart` and per-stage latency breakdown. [2 hrs]

□ **Task:** Assemble the full dashboard layout: two-column grid for Panels 1-2, full-width for Panels 3-6. Add page header “Electoral Equilibrium — Analyst Dashboard” with live estimate count from DuckDB. Test all six panels with real pipeline data. Verify panel filters work. [1 hrs]

Day 3 (Wed — Feedback, Integration Tests & Replication Package)

Task: Schedule and hold an overdue supervision check-in with Prof. Espinosa (last check-in was Week 2; major methodology has changed unreviewed since). Walk in with a written agenda, priority-ordered: (1) the synthetic data augmentation pipeline (DeepSeek→Gemini→human review, 559 LLM-generated records merged into the 1,136-record training set) — is training a forecaster on LLM-labeled data methodologically defensible, and how must it be described/caveated in the methods section? Raise the 647-record unreviewed queue and the shared-blind-spot risk yourself; (2) the definition of `target=0.5066` (drives win_prob saturation); (3) the win_probability CI meaning (sampling error vs Σ_{Δ} -propagated uncertainty); (4) ratify the NEP-weighted baseline change (μ_{eff} 0.716→0.605); (5) confirm the model-vs-market framing. Capture every answer in `DECISIONS.md` and update the

corresponding flag-log items to “resolved/deferred per Espinosa *jdate*”. Several correctness flags CANNOT close without this meeting — it is a prerequisite, not a formality. Severity: blocking for the SRP deliverable. [3 hrs]

Task: PROVISIONAL pending Espinosa check-in (last check-in Week 2). Working assumptions used to keep development unblocked — each MUST be confirmed or corrected at the meeting; do NOT treat as settled:

- `target=0.5066` assumed to mean national two-party popular-vote share needed for an EC win. [PROVISIONAL — likely too low; Dems need ~ 0.52]
- `win_probability` CI assumed to be Monte Carlo sampling error, explicitly NOT covariance-propagated. [PROVISIONAL — methods section must state whichever he confirms]
- Synthetic-data training accepted as-is pending his ruling on whether LLM-labeled data is methodologically admissible. [PROVISIONAL — highest-risk assumption]

Every number/figure produced under these is marked PROVISIONAL in `DECISIONS.md` until the meeting. The flag-log items stay OPEN — these defaults do not close them. [2 hrs]

☐ **Task:** Write `tests/test_integration_end_to_end.py`: run the full pipeline on the smoke config from a clean state. Assert all stage artifact files exist at the expected paths, parse into their typed payload classes without error, and pass `validate()`. Assert the artifact envelope contains `stage`, `run_key`, `metadata`, `data` keys. [2 hrs]

☐ **Task:** Write `tests/test_determinism_full_run.py`: run the pipeline twice with the same config and seed writing to two separate output directories. For each stage, load both artifacts, parse the `data` payload dict, and assert field-by-field equality for all non-path fields. Log any non-deterministic fields found. [2 hrs]

☐ **Task:** Write `tests/test_artifact_schemas.py`: for each artifact produced by the smoke run, assert the envelope keys exist with correct types (`stage` is str, `metadata` is dict, etc.), and assert all required payload fields declared in the artifact classes are present in the `data` dict. [2 hrs]

☐ **Task:** Run the full test suite on a clean virtual environment from a fresh git clone. Fix any import errors, missing `__init__.py` files, hard-coded absolute paths, or undeclared dependencies until the full suite passes on a clean machine state. [2 hrs]

☐ **Task:** Run the complete pipeline on real data with `configs/paper_baseline.json`. Record total runtime per stage. Commit all stage artifacts to `artifacts/paper_baseline/`. Tag `v1.0-paper-baseline`. These artifacts must not be modified after this point — they are the paper’s ground truth. [2 hrs]

☐ **Task:** Write `README.md` “How to Reproduce This Run” section and

`scripts/verify_artifacts.py`. README must include exact Python version, step-by-step commands, HPC setup, expected runtimes, and how to verify outputs. The script loads paper-baseline and fresh artifacts side-by-side and prints a pass/fail report. [2 hrs]

□ **Task:** Write the **Methodology** section (target: 1,800 words). Structure: (1) *Data Sources* — describe the five survey sources and the social media collection across four platforms; introduce the baseline-adjustment procedure for selection bias. (2) *Voter Equilibrium Framework* — state the portfolio analogy formally with the QP from the development plan; define $V_{eq} = 0.535$ and explain the 3% margin-of-error rationale. (3) *Two-Stage NLP Pipeline* — describe RoBERTa as feature engineering layer, explain the fine-tuning dataset construction including social media elasticity features, state the QLoRA hyperparameters in a table. (4) *Constrained Decoding* — one paragraph explaining why standard sampling is insufficient for the web app and how outlines guarantees schema validity. [4 hrs]

□ **Task:** Write the **Results** section (target: 1,200 words). Structure: (1) *Baseline Portfolio* — report the optimized bloc weights and cross-validate against Prof. Espinosa’s documented thresholds (Evangelical 24%, etc.); include the LOCO validation accuracy table. (2) *Baseline Comparison* — present the three-way MAE table (news-only, social-only, unified LLM); highlight which blocs benefit most from the social media signal. (3) *Shock Simulations* — present win probability and bloc-weight shifts for 3–4 representative historical shocks (one per category); include the sensitivity analysis plot for the Ayatollah event. (4) *Platform Correlations* — one paragraph reporting the per-platform correlation with 14-day lagged favorability polls. [4 hrs]

Day 4 (Thu — Paper Completion & Poster)

□ **Task:** Write the **Introduction** (target: 600 words). Open with a concrete example: the Ayatollah shock and its measurable effect on evangelical favorability. State the three gaps in existing work: (i) polling aggregators are static; (ii) advanced systems are enterprise-only; (iii) religious and demographic granularity is underused. State the three contributions: (i) the voter equilibrium framework; (ii) the two-stage RoBERTa-to-LLM pipeline with social media augmentation; (iii) the public web application. End with a roadmap of the paper. [2 hrs]

□ **Task:** Write the **Literature Review** (target: 700 words). Four subsections: (1) *Electoral Forecasting* — FiveThirtyEight, Economist model, prediction markets; their shared limitation as polling aggregators. (2) *Religion and Race in Presidential Politics* — cite Prof. Espinosa’s series (all five books, key articles); the Evangelical threshold, Latino Pentecostal patterns, Catholic swing behavior. (3) *NLP in Political Science* — sentiment analysis for event detection, existing social media political forecasting work. (4) *Portfolio Theory in Non-Financial Contexts* — precedents for applying mean-variance optimization outside finance. [2 hrs]

□ **Task:** Write the **Discussion and Limitations** (500 words) and **Conclusion** (300 words). Discussion must address: (i) platform selection bias and the baseline-adjustment mitigation; (ii) small fine-tuning dataset and why QLoRA regularizes against overfitting; (iii) the model should display uncertainty prominently and must not be used for campaign strategy. Conclusion: summarize contributions, state the 2026 midterm validation plan, note the open-source release. [2 hrs]

□ **Task:** Design the conference poster ($48'' \times 36''$) in Canva. Sections: (i) title block with authors and CMC; (ii) pipeline diagram (the full flow from data to web app); (iii) the baseline comparison table (`roberta_news_only` vs. `roberta_social_only` vs. `llm_unified` MAE per bloc); (iv) a `CoalitionChart` screenshot for the Ayatollah event; (v) the win gauge for one Electoral Surprise shock; (vi) a dashboard screenshot (Panel 1 coverage matrix or Panel 6 audit log); (vii) QR code linking to the live Modal + Vercel URL. Export as PDF and PNG. Share with Prof. Espinosa for any domain input before finalizing. [2 hrs]

Day 5 (Fri — Deployment, Final Submission & Retrospective)

□ **Task:** Incorporate any domain feedback from the Week 8 supervision check-in. Finalize paper and poster as PDFs. Submit both to the SRP committee per their submission instructions. [2 hrs]

□ **Task:** Verify the Modal + Vercel deployment is stable: run 20 back-to-back inference calls from the public URL, confirm all return valid JSON, confirm mean latency is under 5 seconds (Modal cold start included). If latency is too high, switch Modal to a `keep_warm=1` container to eliminate cold starts. Document final latency numbers in `docs/latency_benchmark.md`. [1 hrs]

□ **Task:** Verify the CMC HPC hosting scaffold: `sbatch deploy/hpc_vllm.sh` on CMC HPC and confirm the vLLM server starts and the health check passes on the HPC-internal URL. Do not expose externally yet — this is a dry run to confirm the scaffold works. Log the internal URL and SLURM job ID in `docs/hpc_deployment_notes.md` for future activation. [1 hrs]

□ **Task:** Pin all dependency versions: run `pip freeze > requirements.txt` in the activated venv. Update `pyproject.toml` to match. Commit. Confirm a fresh install from `requirements.txt` passes the full test suite. [1 hrs]

□ **Task:** Write `CONTINUOUS_UPDATE.md` — the post-SRP maintenance specification. This document ensures the model does not become stale after the project ends. Required sections:

1. **Nightly data ingestion.** A launchd plist on the **Intel Mac** triggers a Prefect flow at 2am using `pipeline_mode="continuous"`: runs `BlueSkyCollector` and `ApifyCollector` for any shock events in the last 7 days; runs `sample_archives.py` on any new archive data; runs `clean_with_llm.py` on the new sample; scores new posts via RoBERTa; appends to `data/finetune/new_events.jsonl`. The Intel Mac is the right machine for this: it is

always on, already running collectors, and can submit USC Laguna fine-tuning jobs when the threshold is reached without any human intervention. Include the `launchd` plist and the `just continuous` Justfile target.

2. **Fine-tuning trigger.** When `new_events.jsonl` reaches 10+ new shock events, submit a new USC Laguna fine-tuning job. Archive the current adapter with a timestamp before replacing it.
3. **Voter panel refresh.** Annual check: ARDA, GSS, Pew, and Gallup all release updated data. When new data is available, re-run the Week 2 pipeline. The win threshold $V_{eq}^{(P)}$ and constraint bounds update automatically via `build_constraint_spec`.
4. **Prediction market calibration.** Quarterly: query `audit.duckdb` for resolved shock events; compare `win_probability` predictions against actual market outcomes. Document any systematic divergence and adjust fine-tuning example weights accordingly.
5. **How to trigger a manual run.** Commands for `just continuous`, `just train`, and `just deploy`.

[2 hrs]

□ **Task:** Create `configs/continuous.json`: the configuration file for nightly incremental runs. Set `pipeline_mode: "continuous"`, `lookback_days: 7` (only collect shocks from the last 7 days), `skip_voter_panel_rebuild: true` (voter panel only rebuilt annually), `finetune_trigger_threshold: 10` (trigger retraining after 10 new shock events). Add `just continuous` to the Justfile: `prefect run pipeline.py --config configs/continuous.json`. [1 hrs]

□ **Task:** Final PR: `feature/week8-hardening-deliverables`. This PR must include: (i) all integration and hardening tests, (ii) the frozen paper-baseline artifacts, (iii) `requirements.txt` with pinned versions, (iv) `scripts/verify_artifacts.py`, (v) the complete `README.md`, (vi) the Justfile with all targets verified to work on a clean checkout including `just continuous`, (vii) `DOMAIN.md` reviewed and complete, (viii) `CONTINUOUS_UPDATE.md` complete. Merge to `main`. Tag as `v1.0`. [1 hrs]

□ **Task:** Write `docs/retrospective.md`: a one-page honest assessment covering (i) what went as planned, (ii) what took longer than expected and why (likely: API approvals, HPC queue wait times, data cleaning), (iii) what you would do differently with a 10-week timeline, (iv) the concrete publication plan (target journal, submission timeline), (v) the 2026 midterm validation strategy (which shocks to monitor, how to update the model with new data). [2 hrs]

□ **Task:** Final supervision check-in with Prof. Espinosa: project wrap-up. Share the submitted paper and poster. Get domain sign-off on the framing and discuss the 2026 midterm validation plan. Agree on publication timeline and the 2–3 most important follow-up directions. [2 hrs]

9 Week 8 — Debugging & Hardening

This week consolidates debugging and hardening deferred during the Week 5–6 feature sprints. Tasks are ordered by whether they block the SRP deliverable (training must complete; model must be directionally correct) versus polish. Infrastructure and training pipeline fixes are prerequisites; model quality and data pipeline tasks can proceed in parallel once training is unblocked.

Standing Process — Flag Capture

Task: Maintain a running flag log for the remainder of the project. Every time a development session surfaces a bug, an unverified assumption, an “it should work but isn’t confirmed”, a “decide with Prof. Espinosa”, or a “this contradicts that” — it gets appended here as a checkbox item with a one-line description and a severity (blocking / correctness / hardening / polish). Nothing gets fixed silently and nothing gets forgotten by living only in a chat transcript. New flags append to the end; resolved flags are checked off, not deleted, so the audit trail survives. [1 hrs]

Task: Flag-log entries captured so far (append below as new ones arise):

- ☐ **[correctness]** Meaning of `target` (0.5066) undefined — popular vote? EC threshold? Drives `win_prob=1.0` saturation. Resolve with Espinosa.
- ☒ **[blocking — RESOLVED 2026-06-24 LIVE]** Newsom directional error confirmed fixed on production Modal A10 GPU with v2 adapter + seeded constrained decoding: `white=slight_neg`, `latino=slight_neg`, `evangelical=mod_neg`, `secular=slight_pos`. Direction differentiates correctly by bloc. Latino is no longer boundary-unstable under the seeded path. MVP directional validation PASSED. Paper-grade re-check needed to confirm these exact values are stable across ≥ 3 seeded runs before citing in a results table.
- ☐ **[correctness]** Party-neutral preset text + separate `party` field: confirm model reads the field and flips deltas by party.
- ☐ **[correctness]** ShockNarrative (pre-rebalance sentiment) vs CoalitionChart (post-rebalance weights) must not read as contradictory.
- ☐ **[blocking]** Laguna bitsandbytes hang — bf16 fallback written but never confirmed working.
- ☐ **[hardening]** Hopper 50 GB quota: HF cache, checkpoint bloat, 3.8 GB `.git` with committed weights.
- ☐ **[correctness — CONFIRMED OPEN 2026-06-24]** Production covariance is the identity/diagonal fallback, NOT Ledoit-Wolf. Confirmed live: `win_prob CI [0.653, 0.668]` is MC sampling error around diagonal cov. OK for MVP; blocks paper-grade results. Fix: load real panel data so `run_ilr_montecarlo` uses the 20-cycle Ledoit-Wolf Σ_Δ rather than the near-zero fallback.
- ☐ **[correctness]** No artifact may serialize `Infinity/NaN` — shared sanitizer + test.

- **[correctness]** Apply + verify all six Copilot PR fixes (binomial bootstrap, field comments, dual artifact filenames $\times 3$, fallback method label).
- **[hardening]** Green test suite on a clean checkout after mid-development schema churn.
- **[hardening]** Cross-machine hardcoded paths audit (Mac / Laguna / Hopper).
- **[hardening $\rightarrow$ possibly blocking]** 647-record Gemini human-review queue is EXCLUDED from the current 1,136-record training set (577 original + 559 Gemini-approved) — cautious default. Gate: validate Newsom direction on the 1,136-set model. If corrected $\Rightarrow$ queue is quality-improvement only (adjudicate when convenient). If NOT corrected $\Rightarrow$ queue is blocking — some held-out records may be required for directional correctness and must be adjudicated and merged before the model is paper-ready.
- **[polish]** candidates.jsonl overwrite guard; DeepSeek truncation handling; malformed-JSONL defensive parsing.
- **[hardening]** Backfill or document 60–66 missing (cycle, bloc) survey panel cells.
- **[correctness]** Data-provenance audit: every “real” number traces to a documented source.
- **[correctness]** CoalitionChart dual-axis overlay: loyalty and weight occupy distinct visual ranges on a shared $[0,1]$ axis — verify no per-bloc collision on real shock data before shipping as the SRP headline figure.
- **[correctness]** WinGauge model-vs-market juxtaposition: model answers a counterfactual-conditional question, market answers an unconditional-current question — UI must label them as different, and must flag when the matched contract is the generic election market (not event-conditioned). Confirm framing with Espinosa.
- **[correctness (reporting)]** SSE streaming “30% latency improvement” claim is wrong — stages are sequential so streaming improves time-to-first-content only, not total round-trip. Update devplan/docs to claim perceived-latency improvement, not throughput speedup.
- **[correctness]** Modal audit DuckDB is ephemeral — container restarts silently drop all audit rows. Mount on a Modal Volume or external DB; confirm with a deploy-restart-query test.
- **[correctness — Espinosa-grade]** Bias/symmetry of the model across parties not yet formally validated. Matched scandal pairs show: race deltas near-symmetric (Rep aa -0.07 vs Dem -0.035 ; remaining blocs equal); religion deltas differ by coalition shape (Dem scandal: evangelical/prot $+0.012$; Rep scandal: evangelical/catholic -0.07) — signs are coherent (scandal hits own base hardest), not obvious anti-R bias. BUT: `delta_eff` inverts oddly (Dem scandal $\rightarrow$ Dem $\Delta_{\text{eff}} = -0.0$; Rep scandal $\rightarrow$ Rep $\Delta_{\text{eff}} = +0.03$) — possible per-party aggregate calibration bug that could drive win-prob asymmetry. Required: run 5–10 matched event-type pairs, measure symmetry formally (e.g. sign-parity and magnitude ratio across parties) before any claim that the model is “bipartisan.” This is a methodology question for the Espinosa check-in — a model presented as party-neutral must demonstrate it empirically.
- **[blocking — PUBLIC DEPLOY + LIVE COST]** `min_containers=1` is deployed (warm A10, fixes CORS/150s timeout root cause) but costs $\sim \$1/\text{hr}$ 24/7 *right now*. Must:

(a) set Modal billing hard limit immediately; (b) add per-IP rate limit and concurrency cap before sharing the URL; (c) scale to `min_containers=0` after the demo window closes. This is no longer hypothetical — it is live spend. **Background:** Modal enforces a 150s web-request timeout; requests over it get a 303 redirect that strips CORS headers. Cold start triggered a 14 GB model re-download (`snapshot_download` used `local_dir` not HF cache format, so `from_pretrained` re-fetched every time, ~ 3 min). Fixed: correct `cache_dir` format + `min_containers=1`.

- **[correctness — CONFIRMED OPEN 2026-06-24]** win_probability CI [0.653, 0.668] observed live — a 1.5 pp width that is MC sampling error around the diagonal fallback cov, not covariance-propagated uncertainty. Will widen meaningfully once Ledoit-Wolf Σ_Δ is active (ties to covariance-source flag above). Decide with Espinosa: (a) Wishart-propagated outer loop, or (b) explicitly redefine as sampling-error bounds and document scope. Do not present this CI as the methodology-section interval until the covariance source is fixed and the semantic choice is documented.
- **[correctness — paper deliverable]** Synthetic data ablation study not tracked: DECISIONS.md requires training two models — hand-authored-only vs augmented — and reporting both MAEs before the synthetic rows may be claimed as beneficial in the paper. If the augmented model does not improve on hand-authored-only, the synthetic rows must be dropped. This is a required paper section, not optional.
- **[correctness]** `bin_uncertainty.json` σ_b values are still unpopulated (open item in DECISIONS.md). The Logistic-Normal MC uses per-bin empirical σ_b ; if this file holds defaults/zeros the CI width is wrong. Run `generate_synthetic.py` to populate before the paper-baseline run.
- **[correctness — paper claim]** LIWC cross-validation of the RoBERTa scorer is blocked (no tweet-ID join column in the SAV file). DECISIONS.md defers the alternative (100-post Spearman) to post-SRP. The methodology section must NOT claim LIWC validation occurred; it must accurately describe what validation was done (score distribution variance check only).
- **[correctness/legal]** Foster & Rathlin `metoo_liwc` dataset is CC BY-NC-SA 4.0 (non-commercial only) — flagged in DECISIONS.md for startup/commercial review. Confirm the SRP publication, any web demo, and any downstream use does not violate the non-commercial clause before submission.
- **[correctness]** NEP-weighted baseline change (μ_{eff} 0.716 $\rightarrow$ 0.605) is listed in the Espinosa meeting agenda (item 4) as “ratify” but does not appear as a decided entry in DECISIONS.md. Either document it as an implemented decision or confirm with Espinosa that it is correct and then document it.
- ☒ **[blocking — deploy — RESOLVED 2026-06-23]** Constrained decoding fixed end-to-end: `pydantic.create_model` + str-Enum + `outlines.models.Transformers` wrap + `use_cache=False` + Enum `.value` unwrap. No fallback. Schema guarantee restored. Deploy blocker cleared.
- ☒ **[correctness — paper — RESOLVED 2026-06-23]** Decode non-determinism closed. Generation seeded (`seed` param threaded through `predict_delta_bins` $\rightarrow$

`_predict_constrained`, default 42, `torch.manual_seed + rng`). Confirmed reproducible: identical output across two runs. Inference path fully working: trained adapter + constrained decoding + seeded + clean str tokens. **All MVP code blockers closed.** Remaining paper obligation: headline-event validation tables must be produced under the seeded constrained path; boundary-straddling blocs (latino on Newsom) must be noted in the paper as carrying uncertainty the MAE metric does not capture.

- **[correctness]** Python 3.10+ union syntax (`X | None`, `X | Y`) crashes at import on local 3.9 dev env — already broke FastAPI server boot (`get_audit`, `str | None`). Choose one fix: (a) convert to `typing.Optional/Union` throughout `electoral/`; (b) add `from __future__ import annotations` to every affected module (verify no runtime annotation introspection); or (c) bump local dev to 3.11 to match HPC/paper-baseline target. Clean-checkout test on 3.9 will keep surfacing these until resolved.
- **[blocking — deliverable]** Methodology and Results sections are scaffolded, not written — 39 TODO/PROVISIONAL placeholders, each tied to a real output (Espinosa decision, paper-baseline run, or model validation). None may be filled from memory or assumption. Submission gate: `grep -c 'TODO\|PROVISIONAL'` on the paper must return 0. Load-bearing holes: contested V_{eq} (0.535 vs 0.5066), synthetic-data admissibility, Newsom directional validation.
- **[blocking — deliverable]** Do NOT tag `v1.0-paper-baseline` until: (a) retrained adapter validated to fix the Newsom all-minorities-positive error; (b) Espinosa-confirmed (or explicitly-provisional) `target` and CI definitions; (c) green clean-checkout test suite; (d) production covariance confirmed as Ledoit-Wolf, not the diagonal fallback. Tagging early freezes known-wrong numbers as immutable ground truth.
- **[correctness/security — blocking for public deploy]** Dashboard page auth (`DASHBOARD_PASSWORD`) does not protect the FastAPI backend routes (`/api/coverage`, `/api/sentiment-dist`, `/api/bio-coverage`, `/api/audit`) on the Modal URL — anyone can call them directly and read panel data and the full audit log. Add a shared bearer token / signed header checked at the FastAPI boundary before going public.
- **[security — blocking for public deploy]** `/api/audit` search term is interpolated into SQL (`LIKE %search%` via f-string/concatenation) — SQL injection on a public endpoint hitting the audit store. Use a DuckDB placeholder (`ILIKE ?` with wildcards in the bound parameter). Clamp `limit`. Audit every other DuckDB query for the same pattern.
- ☒ **[correctness — RESOLVED 2026-06-29]** CoalitionChart rendered only the `white` bloc (4 of 5 bars invisible). Root cause: Recharts 3 with `layout="vertical"` sets `props.width` to 0 during initial render for most bars; the old guard `if (width <= 0) return <g/>` silently dropped them. Fix: derive `plotW` from `background.width` (`offset.width`, always the full chart-plot pixel span) and compute `barW = sv * plotW`. All five race-bloc bars plus the translucent weight overlay now render correctly. TypeScript type-check and production build both pass. Confirmed in `webapp/components/CoalitionChart.tsx`.
- **[polish]** Remove the `console.log("[BarShape]", ...)` debug statement in `CoalitionChart.tsx` (gated by `NODE_ENV === "development"` but still noise). Remove once user confirms all five blocs render correctly in the browser.

- **[correctness — optimizer]** Coalition weights $\tilde{w}$ were demographically unconstrained: the optimizer concentrated weight on the highest-loyalty bloc (`african_american` $\tilde{w} \approx 0.50$ vs ~ 0.19 electorate share), producing unrealistic coalitions and inflating win probabilities. FIX IN PROGRESS: added per-bloc bounds $w_0[i] \cdot \text{LOWER_MULT} \leq w[i] \leq w_0[i] \cdot \text{UPPER_MULT}$ (provisional $0.5 \times -1.5 \times$ of NEP baseline share in `electoral/optimization/cvx.py`). Bounds are PROVISIONAL — calibration is an Espinosa decision. Verify some shocks now correctly go infeasible (a feature, not a bug).
- **[correctness — Espinosa]** Win-probability inflation: with unconstrained weights essentially all events cleared target (e.g. “MET +16 pp”, win-prob 0.66 for a scandal). Confirm after the weight-bound fix that win-probs attenuate to realistic levels. Ties to the open `target=0.5066`-meaning and diagonal-covariance flags.
- **[correctness — simulation]** Per-bloc weight percentile (p5–p95) panel renders degenerate: percentiles span ~ 0 –0.88 with mass piled near 0, an artifact of (a) diagonal-covariance fallback and (b) unconstrained optimizer. Recheck after weight-bound fix; if still degenerate, either wire real Ledoit-Wolf Σ_Δ or remove the panel for the presentation and note as future work.
- **[correctness — display]** ShockNarrative has no `neutral` bucket: blocs with `delta_bin=="neutral"` are sorted into hurt/help, making output appear more polarized than the model’s actual predictions (which include many neutrals). Add a third “little measurable effect on:” bucket.
- **[polish — display]** CoalitionChart conflated two distinct quantities ($\tilde{\mu}$ loyalty and $\tilde{w}$ coalition emphasis) in one overlay; users misread $\tilde{w}$ as population share. FIX IN PROGRESS: split into two panels (loyalty shift / coalition emphasis) + add an explainer panel defining each term + relabel $\tilde{w}$ as “coalition emphasis (strategic weighting), not population share.”
- **[polish — display]** Recharts v3 passes `width=0` to custom bar shape during initial layout, tripping the `width<=0` guard; only one bloc’s bar was visible. Partial fix via `background.width` (see RESOLVED entry above). Splitting into two single-quantity panels sidesteps the fragile overlay entirely; verify all 5 blocs render in both panels post-split.
- ☒ **[polish — DONE]** Removed market-prior / model-vs-market module (“No active market”, Polymarket/Manifold comparison): almost always empty, and when populated compared the model’s conditional $P(\text{win} \mid \text{shock})$ against the market’s unconditional $P(\text{win} \mid \text{current})$ — different questions, misleading juxtaposition. Removing it also eliminated `/api/market-prior` CORS errors.
- **[hardening — deploy/cost]** Modal cost control: `min_containers=0`, `scaledown_window=60`, `cache_dir` baked into the image (cold start 23.7s, no 14 GB re-download). HARD CONSTRAINT: keep Modal under \$15/month. NEVER set `min_containers>0` except during an active demo window, then revert. Manual task: set Modal dashboard billing cap. Run all batch/validation work on Hopper (free); Modal only for the live demo URL.
- **[correctness — validation — HIGH PRIORITY]** No systematic symmetry/valence validation has been run. Individual outputs are untrustworthy without it (cannot distinguish “counterintuitive-but-correct” from “biased or pattern-matching party labels”). RE-

REQUIRED before presentation: (1) *symmetry test* — matched event pairs across event types, measure whether the two parties are treated symmetrically; (2) *valence test* — events with known unambiguous direction, confirm aggregate Δ_{eff} moves correctly. Run on Hopper (free). Required paper/presentation deliverable regardless of outcome.

- ☒ [decision — REJECTED] Considered (a) migrating hosting from Modal to Hopper and (b) using Grok 2 as the base model or API output-corrector. Both REJECTED: Hopper cannot host a persistent public URL (job scheduler, no inbound routing); Grok 2 is 500 GB/8-GPU (won't fit), and its Community License prohibits using it to train or improve other AI models, invalidating the synthetic-data and corrector use cases. Grok-as-EVALUATOR (score outputs, don't alter them) is acceptable; Grok-as-CORRECTOR is not (unreproducible, replaces the model's own judgment, license violation).
- ☐ [context — one week to presentation] Priority order: (1) optimizer weight-bound correctness fix; (2) symmetry/valence validation on Hopper; (3) display fixes (neutral bucket, two-panel split, explainer); (4) overdue Espinosa meeting on **target** meaning + covariance + weight-bound calibration.

The SRP is not “done” until every box here is checked or has a written, Espinosa-approved deferral. [2 hrs]

HPC & Infrastructure Task: Resolve USC Laguna `bitsandbytes` hang. Laguna GPU nodes (both c22 and c23) run kernel 4.18.0, below the 5.5.0 minimum. QLoRA 4-bit quantization via `bitsandbytes` hangs indefinitely at training step 0 on every Laguna node. Implement and TEST the `ELECTORAL_NO_4BIT=1` bf16 fallback path in `trainer.py` (skip `BitsAndBytesConfig`, load in `bfloat16` — L40S 46 GB fits 7B bf16 + LoRA easily). The fallback was written but never confirmed working because the env var did not propagate to the trainer in the last attempt. Verify the bf16 branch is actually taken (grep the log for absence of the `bitsandbytes` 4-bit loading line) and that training advances past step 0. [3 hrs]

Task: Fix Hopper disk quota management. The 50 GB home quota is repeatedly exhausted. Root causes: (i) 14 GB HuggingFace Mistral-7B cache in `~/.cache/huggingface`, (ii) intermediate training checkpoints accumulating (~13 MB each, every 113 steps), (iii) a bloated 3.8 GB `.git` directory containing committed model weights. Fix all three: (a) set `HF_HOME` and `TRANSFORMERS_CACHE` to node-local `/tmp` during runs, (b) add `save_total_limit=1` to `TrainingArguments` in `trainer.py`, (c) run `git gc` / filter the `.git` history to purge committed weights. [3 hrs]

Task: Purge model weights from git history. The `.git` directory is 3.8 GB because model safetensors were accidentally committed before `models/` was gitignored. Use `git filter-repo` or BFG to remove `models/` and `checkpoints/` from all history, then force-push. Coordinate timing so Hopper and Laguna clones are re-cloned cleanly afterward. Verify `.git` drops below 50 MB. [2 hrs]

Task: Document the cross-HPC dependency matrix. The working dependency set differs by

machine and must be pinned: `outlines==0.0.37`, `pyairports`, `numpy<2.0`, `scipy<1.14`, `cvxpy==1.3.4`, `transformers==4.46.3`. The `outlines/cvxpy/numpy` versions conflict (`cvxpy>=1.7.5` wants `numpy>=2.0` but `outlines 0.0.37` wants `numpy<2.0`). Document the resolved pinned set in a `requirements-hpc.txt` and in `DECISIONS.md`, noting Hopper requires module load `cuda/13.2` and `LD_LIBRARY_PATH=/hopper/software/cuda/13.2/lib64` in every `srun`. [2 hrs]

Task: Persist the audit DuckDB across Modal container restarts. `data/audit.duckdb` is a local file; Modal’s container filesystem is ephemeral, so audit rows are silently lost on every cold start / redeploy. Mount the DuckDB file on a Modal Volume (or move audit storage to an external persistent DB). Until then the audit log works locally but is decorative in production — confirm persistence with a deploy-restart-query test (log an estimate, force a container restart, confirm the row survives). Severity: correctness — an audit trail that doesn’t persist is not an audit trail. [2 hrs]

Training Pipeline **Task:** Resolve `eval_mae=inf` during training. Both `r=16` and `r=32` training runs report `eval_mae=inf` because the 4-bit model’s in-loop evaluation is broken. Currently MAE is measured post-hoc via `run_baseline_comparison.py`. Either fix the in-loop eval to produce a real number or formally remove the broken eval from the training loop and rely on the post-hoc script. Replace any `Infinity` written to `trainer_state.json` with `null` (JSON-invalid `Infinity` already caused a downstream parse bug, patched once — verify `trainer.py` no longer emits it). [2 hrs]

Task: Fix `outlines` constrained-decoding latency. `ShockEstimator.estimate()` takes ~12s per call (p95 well above the 3s target) because `outlines 0.0.37` FSM-based constrained decoding processes ~200 tokens one-at-a-time. Documented in `docs/latency_benchmark.md` as deferred. Investigate `vLLM guided_json` decoding (expected 2–4s) or a distilled smaller model. This blocks any interactive/web use of the endpoint. [4 hrs]

Task: Fix the KV-cache / `use_cache` workaround. `ShockEstimator` currently sets `model.config.use_cache=False` to avoid a `NoneType.device` crash in `outlines’ reorder_kv_cache` under PEFT models. This disables KV caching and worsens latency. Find a proper fix that keeps caching enabled. [3 hrs]

Task: Audit the intensity-clipping behavior. `estimate()` clips deltas to `[-0.15, 0.15]` *after* intensity scaling to satisfy `ShockResponseData` validation. At intensity = 1.5–2.0 this clips many blocs to the boundary, flattening the sensitivity analysis. Decide whether clipping should happen pre- or post-scaling and document the modeling justification. [2 hrs]

Model Quality **Task:** DONE (MVP): Validate the retrained `r=16` model fixes the directional errors. Pre-augmentation: `slight_pos` for all minority blocs on Newsom 2028 (wrong). After

retraining on the 1,136-record augmented dataset, confirmed on production Modal A10 with v2 adapter + seeded constrained decoding: `white=slight_neg`, `latino=slight_neg`, `evangelical=mod_neg`, `secular=slight_pos`. Direction differentiates correctly by bloc. REMAINING (paper obligation): run ≥ 3 seeded passes under the same adapter/decoding path and produce a headline-event results table. Latino is boundary-unstable (note this in the paper). If the model is wrong on any of the 3 runs, escalate to Espinosa before writing the results section. [3 hrs]

Task: Verify party-neutral event text + explicit party field produces party-appropriate deltas. The webapp uses an authoritative party toggle with party-neutral preset templates (e.g. “the leading presidential candidate”, “the sitting administration”) — the party is passed as a separate field in the event dict, not named in the description. But the model was fine-tuned on synthetic seeds that DO name the party in the text (“Democratic nominee...”, “Republican administration...”). Confirm the model reads the separate `party` field correctly: send the same neutral event text with `party="democrat"` and `party="republican"` and verify the predicted deltas flip in the politically expected direction. If they do NOT (i.e. the model ignores the party field when the text is neutral), the fix is to have the frontend inject the party name into the event text at submit time (template “the {party} candidate”) rather than relying on the field. Decide and document which approach is authoritative. [2 hrs]

Task: Resolve the `win_probability=1.0` saturation. With NEP-weighted w_0 , the Democrat baseline μ_{eff} is 0.605 vs. target 0.5066 — a 0.098 gap that makes `win_probability` saturate at 1.0 across all shock intensities. The diagonal fallback covariance (near-zero variance, used when fewer than 3 panel cycles) produces degenerate CIs of [1.0, 1.0] or [0.0, 0.0]. Confirm the Ledoit-Wolf covariance from 14 real cycle-deltas is actually being used in production (not the fallback), and verify the target value 0.5066 is the correct Electoral College threshold — Democrats are structurally disadvantaged and may need ~ 0.52 – 0.54 popular-vote-equivalent to win the EC. Clarify what “target” represents with Prof. Espinosa. [3 hrs]

Task: Verify the NEP-weighted baseline formula end-to-end. The w_0 computation was changed from a degenerate CVXPY-derived result (`asian= 0`, `latino= 0`, `african_american= 0.51`) to $w_i = (\text{NEP}_i \cdot \mu_i) / \sum_j (\text{NEP}_j \cdot \mu_j)$ using Edison Research 2024 exit-poll voter shares. Confirm `build_baseline_portfolio` uses this formula (not `solve_baseline`) for w_0 , and that the resulting weights (`white= 0.528`, `african_american= 0.189`, `latino= 0.180`, `asian= 0.051`, `other_race= 0.052`) propagate correctly into the optimizer and Monte Carlo. `solve_baseline` should be used *only* for post-shock rebalancing. [2 hrs]

Data Pipeline Task: Fix the `candidates.jsonl` overwrite hazard. The DeepSeek generator opens the output file in write mode, so an accidental second run silently overwrote 1,237 good records with 0 (the run kept=0 because the file already existed at the seed level). Add a guard: refuse to overwrite a non-empty output file without `--force`, or append with dedup. Also reduce per-attempt warning noise (warn only on final failure). [2 hrs]

Task: Fix DeepSeek truncation at scale. Generating 20 records per call truncated mid-record at the 4k token limit (kept=0 for several seeds). Worked around with `BATCH_SIZE=5` and `max_tokens=8000`. Make batch size and token budget configurable and add a post-parse check that warns when a response appears truncated (ends mid-object). [1 hrs]

Task: Handle malformed JSONL in review outputs. 7 of 647 human-queue records are unparseable JSON (Gemini occasionally emits invalid output). All readers must parse defensively (skip-and-count bad lines) rather than crashing on `json.loads`. Audit every place that reads these JSONL files. [1 hrs]

Task: Backfill missing survey panel cells. `build_voter_panel` reports 60–66 missing (cycle, bloc) pairs across ANES/CES/VOTER sources — notably evangelical, secular, jewish, muslim, `other_gender` absent for pre-2004 cycles, and most blocs missing for 2024. The religion strata impute to 0.50 with warnings. Source the missing data or document the imputation as a known limitation with its effect on μ_{religion} . [3 hrs]

Synthetic Review Backlog Task: Adjudicate the 647-record human review queue. Gemini 2.5 Flash flagged 647 of 1,237 synthetic records as `HUMAN_REVIEW` (conservative by design; 579 were auto-approved and already merged, bringing the training set to 1,136). Process `human_review_queue.jsonl` in an Opus session (manual upload during dormant hours): approve, correct, or reject each. The queue includes ~15 records Gemini actually approved but were routed via the 5% spot-check mechanism — these validate that approved records are trustworthy. Merge keepers and retrain. [4 hrs]

Task: Spot-check the merged synthetic set for shared-blindspot errors. Three models (DeepSeek generate, Gemini review, the prior Mistral) can share the same political blind spots — e.g. all assuming a far-left event helps Democrats. Sample 50 high-confidence approved records plus the highest-disagreement tail and review with Prof. Espinosa for systematic directional bias before the labels are trusted as ground truth. [3 hrs]

Task: Add coverage-gap detection to the generator. Review accuracy does not fix coverage gaps — if DeepSeek under-produces a taxonomy cell (e.g. `helps_rep/far_right` institutional events), no amount of review surfaces the absence. Add per-axis-cell counting to `generate_deepseek.py` and oversample underfilled cells in a second pass. [2 hrs]

API & Frontend Contract

Task: Widen the `/estimate` endpoint to return the full artifact bundle. The webapp's `EstimateResponse` type assumes the endpoint returns `shock`, `equilibrium`, and `simulation` together, but `shock_endpoint.py` currently returns only `ShockResponseData`. Extend the POST `/estimate` route to run the full chain (`build_shock_response` → `solve_rebalanced` → `run_ilr_montecarlo`) and return a composite JSON object `{shock, equilibrium,`

`simulation`} matching the `EstimateResponse` TypeScript interface. Add per-stage timing fields (`llm_ms`, `optimizer_ms`, `montecarlo_ms`) so the frontend `AuditEntry` can be populated from a single call. Alternative considered and rejected: three separate endpoints — adds round-trips and state-sync complexity. Keep it one call. [3 hrs]

Task: Establish a single source of truth for the API contract. The TypeScript types in `webapp/lib/types.ts` are hand-mirrored from the Python dataclasses in `electoral/artifacts.py` and will drift. Either (a) generate the TS types from the Pydantic `ShockResponseSchema` via `pydantic-to-typescript` or a JSON Schema export, or (b) add a CI check that fails when the two definitions diverge. Cover `ShockResponseData`, `EquilibriumData`, `SimulationData` (including the `win_probability_low/high` fields added in Week 6), and the `Party/DeltaBin` literal types. Document the chosen approach in `DECISIONS.md`. [3 hrs]

Task: Reconcile the latency budget with the endpoint design. The full `/estimate` chain includes the ~12s outlines constrained decoding (documented in `docs/latency_benchmark.md`), so a single composite call will take ~12–15s end-to-end. The frontend must handle this: add a loading/streaming state, and decide whether Monte Carlo (the fast part) should stream separately from the LLM estimate (the slow part). Cross-reference the Week 8 latency task (`vLLM guided.json`) — if that lands, this budget shrinks to ~3–5s. Until then the UI must not appear hung during the 12s wait. [2 hrs]

Task: Define the `TrainingRun.val_loss` and `mae` null contract. These webapp fields are nullable because in-loop evaluation currently produces `eval_mae=inf` (the 4-bit eval bug). The frontend must render “n/a” gracefully when these are null, and the `TrainingRun` feed must serialize Python `inf/NaN` as JSON `null` (never the invalid `Infinity` literal that already caused one parse bug). Add a serialization guard. [2 hrs]

Task: Populate `CoverageCell` quality flags from the real panel data. The webapp coverage grid expects per-(bloc, cycle) quality of “real” / “imputed” / “missing”, but the panel builder currently only logs missing pairs as warnings (60–66 missing cells, mostly religion strata pre-2004 and most blocs in 2024). Emit a structured coverage artifact from `build_voter_panel` that the endpoint can serve, rather than the frontend guessing. [2 hrs]

Task: DONE: Implement the `/estimate/stream` SSE endpoint in `shock_endpoint.py`. FastAPI `StreamingResponse` with `media_type="text/event-stream"` is deployed and working on the live Modal A10. Named events `deltas`, `equilibrium`, `simulation`, and `done` are emitted in order as each pipeline stage completes. `page.tsx` opens an `EventSource` and populates state via `onEquilibrium/onSimulation` callbacks with Zod v4 schema validation at the SSE boundary. Progressive-render UX confirmed working end-to-end (charts populate incrementally as each stage returns). OPEN: SSE “30% latency improvement” claim is incorrect (see latency flag in the log above — streaming improves time-to-first-content, not total round-trip); update devplan copy accordingly. [3 hrs]

Task: Reconcile the two endpoint shapes. The webapp now depends on BOTH `POST /estimate` (returns the full bundle, validated by `EstimateResponseSchema`) and `GET /estimate/stream` (SSE, incremental). Decide whether both are maintained or the non-streaming one is dropped. If both stay, the Zod `EstimateResponseSchema` should also validate the assembled-from-stream object so the streamed path gets the same runtime safety as the single-shot path. [2 hrs]

Task: Verify `ShockNarrative` and `CoalitionChart` tell the same story. The narrative renders plain-language “hurt/help” sentences from raw `delta.bins` (the model’s pre-rebalance sentiment prediction), while `CoalitionChart` shows both the baseline bars AND the post-optimization rebalanced overlay. These are two different layers — sentiment movement vs. campaign resource reallocation — and must not be read as contradicting each other. Confirm: (i) the narrative explicitly describes direction-of-movement (“standing with”), never final level (“the party is winning X”); (ii) a code comment marks `ShockNarrative` as intentionally pre-rebalance so no one later rewires it to equilibrium weights; (iii) for a test shock, the blocs the narrative says are “hurt” are not simultaneously shown gaining weight in the chart in a way a layperson would read as contradictory — and if they can be (a bloc whose sentiment drops but whose optimal campaign weight rises), add a one-line chart caption explaining the distinction. The plain-language layer is the one a reader takes at face value, so it must be faithful to exactly what the number means. [2 hrs]

Task: Pressure-test the `CoalitionChart` dual-axis overlay on real result data. The chart overlays within-bloc loyalty (μ_i , ~ 0.15 – 0.90) and coalition weight (w_i , ~ 0.03 – 0.30) on one shared $[0,1]$ axis, relying on them occupying distinct ranges to stay visually unambiguous. This breaks when a bloc’s loyalty and weight land near the same value (e.g. White loyalty 0.30 vs weight 0.28) — the two bars overlap and a layperson reads the translucent bar as a loyalty drop, not a different quantity. Render the chart for the 3–4 most consequential real shocks and check for this collision. If it occurs, either (a) split into two stacked sub-charts sharing a Y axis (loyalty panel + weight panel), or (b) move weight to a secondary top X axis with its own scale and label. Do NOT ship the shared axis as the SRP headline figure until this is verified on real numbers. Severity: correctness — the figure must not mislead. [2 hrs]

Task: Make the model-vs-market juxtaposition epistemically honest. The `WinGauge` shows model $P(\text{win} \mid \text{hypothetical shock})$ next to market $P(\text{win} \mid \text{everything currently true})$. These answer DIFFERENT questions — the model is counterfactual-conditional, the market is unconditional-current — and for most hypothetical events no contract is actually conditioned on the event, so “nearest relevant contract” returns the generic election market (the unconditional prior). Presenting “Model 62% / Market 55%” as if they were the same question is misleading. Required: (i) UI copy must label them as different questions, not competing estimates — e.g. “Model: if this happens $\rightarrow$ ” vs “Market (current, unconditional): $\rightarrow$ ”; (ii) when the matched contract is NOT conditioned on the event (the common case), say so explicitly (“nearest market is the general election winner, not this scenario”); (iii) the `aggregator.py` match must return whether the contract is event-conditional or generic, and the UI must surface that. Severity: correctness — this is the headline epistemic display and must not imply a false comparison. Confirm framing with Prof. Espinosa. [2 hrs]

Final Sweep — Ambiguities & Loose Ends

Task: Resolve the meaning of `target`. PARTLY DONE. (1) The VALUE is principled: `configs/party_config.json` defines $V_{eq} = 0.5066$ (Dem) / 0.4934 (Rep) as the EC-adjusted popular-vote share where $P(\text{EC win})=0.5$, derived by `derive_ec_veq()` (logistic fit over 20 cycles). (2) RECONCILED: the hardcoded `V_EQ_DEFAULT` fallback now matches $(0.5066/0.4934)$ instead of the stale $0.521/0.495$. `base.json` already agreed at 0.5066 ; `smoke.json` deliberately keeps 0.535 as a fast-test value. (3) FIXED a real bug: the validators required $0.5 < \text{target} < 0.7$, which rejected the Republican EC V_{eq} ($0.4934 \nless 0.5$) and broke every Republican run — widened to $(0.40, 0.70)$ in `PipelineConfig` / `BaselinePortfolioData` / `EquilibriumData`, with regression tests. NOTE: the earlier hypothesis that 0.5066 is “too low” and causes `win_probability=1.0` saturation was WRONG — the saturation comes from the near-zero DIAGONAL Σ_{Δ} (smoke has no real panel covariance; $\mu_{eff} \approx 0.59 \gg 0.5066$ with tiny variance $\Rightarrow$ every draw wins), tracked by the “production covariance path” task. REMAINING (needs Espinosa, a modeling call not a code fix): is μ_{eff} — a λ -weighted blend over three overlapping strata — a fair proxy for the national two-party popular-vote share that V_{eq} thresholds? Document the approximation in `DECISIONS.md`. [2 hrs]

Task: Apply all six Copilot PR review fixes that were accepted but may not all be committed: (i) Bernoulli-binomial bootstrap in `montecarlo.py` (avoid the $50k \times 500$ index matrix), (ii) corrected `win_probability.low/high` field comments in `artifacts.py` (bootstrap of the estimator, not “across ILR draws”), (iii)+(iv)+(v) `shock.py` must write BOTH legacy `shock_response.json/equilibrium.json` AND per-shock `shock_{id}.json/equilibrium_{id}.json`, restoring the “event” metadata key (fixes `test_stages.py` and `test_shock_kernel.py`), (vi) the equal-weight fallback in `cvx.py` must use `method="equal_weight_fallback"`, not “`cvxpy_dqcp`”. Verify each is committed and its test passes. [3 hrs]

Task: Run the full test suite on a clean checkout and fix every failure. Many tests were written against schemas and artifact filenames that changed mid-development (`SimulationData` gained `win_probability` fields; `shock.py` changed artifact names; `cvx` fallback method label changed). Enumerate failures, fix the code or the test as appropriate, and confirm a green suite. Do not leave any test xfail/skip without a documented reason. [3 hrs]

Task: Reconcile every cross-machine path and config assumption. The repo has hardcoded paths for three environments (M5 Mac `.venv` Py3.9, Laguna scratch, Hopper home). Audit for hardcoded `/Volumes/`, `/Users/juhidamley/`, `/scratch/...`, `/hopper/home/...` that will break when run elsewhere. Move them to config or env vars. Confirm `configs/base.json` vs `smoke.json` `data_path` behavior is documented. [2 hrs]

Task: Confirm the production covariance path. DONE: `run_ilr_montecarlo` now accepts a `cov_delta` argument and uses it as Σ_{Δ} (with PSD repair); the real shock covariance is

threaded through every caller — the SSE API (`_run_mc_thread`), the Monte Carlo CLI, and the pipeline stage (`stages.run_simulations ← pipeline.py`). When `cov_delta` is `None` it falls back to the isotropic diagonal, so existing behavior is preserved (tests in `test_montecarlo.py`). STILL TO DO: ensure the Σ_Δ actually SUPPLIED is the genuine Ledoit-Wolf estimate from 14 real cycle-deltas, not itself the diagonal fallback (the smoke artifact’s covariance is currently $0.02^2 I$ because no panel parquet was present). Add a variance FLOOR so a near-zero empirical Σ_Δ can’t collapse the CI to $[1.0, 1.0]/[0.0, 0.0]$, log which covariance source was used, and add a test asserting the fallback is NOT silently used when real panel data is present. [2 hrs]

Task: Final data-provenance audit. Confirm every “real” number traces to a documented source: NEP 2024 voter shares (Edison Research), ANES/CES panel mu values, the 1,136-record synthetic training set (577 original + 559 Gemini-approved, with the 647-record human-review queue either adjudicated or explicitly marked as deferred). No silently-imputed value should be presentable as empirical. Produce a one-page provenance table for the SRP appendix. [3 hrs]

Task: Verify no JSON artifact can contain Infinity/NaN. The `eval_mae=inf` bug already produced one invalid-JSON parse failure. DONE: a shared `sanitize_floats()` sanitizer in `electoral/core/io.py` (recursively maps `inf/-inf/nan → null`), tested in `tests/test_io.py::TestSanitizeFloats`, is now applied at EVERY JSON emission point we control — `write_json/write_artifact`, the Monte Carlo CLI (`simulation/montecarlo.py`), the SSE frames (`api/shock_endpoint.py`), and the fine-tune record serializer (`llm/trainer.py`). REMAINING: `trainer_state.json` is written by the HuggingFace Trainer itself (not our code), so a non-finite `best_metric` there isn’t caught by our sanitizer. Close it with a `TrainerCallback` that sanitizes the metrics dict (or post-process `trainer_state.json` after training), and stop `_eval_mae` from propagating a raw `inf` into the saved state. [2 hrs]

Task: DONE: Enforce per-bloc weight bounds inside the optimizer instead of after it. `solve_dqcp` (`electoral/optimization/dqcp.py`) now enforces the `ConstraintSpec` bounds EXACTLY inside the SOCP via the linear pair `z_i >= lo_i*s / z_i <= hi_i*s` (added as `cp.multiply(lo, s)` constraints), replacing the old post-solve clip+renorm that could push a weight back outside its bound. Also fixed a latent attribute bug (the old code read `spec.lower_bounds/upper_bounds`; the real fields are `spec.lower/spec.upper`). Tests added in `tests/test_optimization.py` assert the upper cap and lower floor hold for specs the unconstrained optimum violates. Behavior-preserving for current runs (no caller passes a spec yet, so the spec branch was dead code). [3 hrs]

Task: DONE: Reconcile the religion/gender basis in μ_{eff} . The post-shock optimizer and Monte Carlo no longer use a flat neutral 0.5 for the fixed strata. A shared `dqcp.compute_fixed_loyalty()` computes the REAL religion+gender contribution $\lambda_2 \cdot \overline{\mu_{\text{rel}}} + \lambda_3 \cdot \overline{\mu_{\text{gen}}}$ from the baseline loyalties shifted by the shock’s religion/gender deltas (post-shock-shifted, consistent with how race uses $\tilde{\mu}$). The kernel passes it (from the baseline artifact); the API passes it (from nominal priors `_NOMINAL_MU_RELIGION/GENDER`);

`run_ilr_montecarlo` DERIVES it from `mu_eff_shifted` so the MC’s win check always matches the optimizer. `solve_rebalanced` / `run_ilr_montecarlo` take an opt-in `fixed_loyalty` (None → old neutral, behavior-preserving). Smoke impact: μ_{eff} 0.5959→0.5900, weights shift ≤ 0.008 . Tests in `test_optimization.py` / `test_montecarlo.py`. STILL SEPARATE (do not conflate): (a) the `v_R/g_G` within- stratum weights are still equal ($1/N$) until `raking.py` supplies real panel marginals; (b) the *value* of `V_eq/target` (EC vs popular-vote definition) is the open “resolve target” task — the basis fix here does NOT change it. Re-freeze the baseline once both are settled, and document in DECISIONS.md. [3 hrs]

Task: DONE: Consolidate the two duplicate optimizer code paths. There used to be two SOCP formulations that solved DIFFERENT problems: the kernel’s `cvx.solve_rebalanced` compared race-only $\mu \cdot w$ to V_{eq} (no λ weighting, but it DID enforce the $[0.05, 0.60]$ bounds), while the API’s wrapper used `dqcp.solve_dqcp` with the full λ -weighted μ_{eff} but NO bounds. Now: `dqcp.solve_dqcp` is the single canonical solver; `cvx.solve_rebalanced` is a thin wrapper that calls it (with a [floor, ceiling] `ConstraintSpec`) and returns an `EquilibriumData`; the FastAPI wrapper delegates to `cvx.solve_rebalanced`. Both paths now use the SAME λ -weighted objective AND the same per-bloc bounds, and `mu_eff_shifted` is populated (was unset on the kernel path). Smoke-input impact: coalition weights move ≤ 0.016 ; `mu_eff_shifted` 0.0→0.5959. Committed `artifacts/smoke/*` still reflect the OLD kernel optimizer and will update on the next real pipeline run (regen needs the LLM; not hand-edited). Religion/gender are still held at neutral 0.5 — switching them to the real loyalties is the remaining “ μ_{eff} basis” task below. [3 hrs]

AI Research Assistant — Quick Reference

Setup: OpenClaw on Raspberry Pi 5 → Telegram/WhatsApp interface → Gemini Pro (default) or Claude Sonnet (@claude prefix). Syncthing shared folder across the three local machines (M5, Intel Mac, Pi).

Machine Allocation at a Glance

Machine	Always running	When needed
M5 MacBook Pro	Development, Claude sessions	MLX fine-tune experiments
Intel MacBook	Bluesky + Apify collectors	FastAPI endpoint in Week 7
	(24/7); nightly pipeline (2am launchd)	
Intel Mac	News scraper (was Windows role), collectors	launchd 2am nightly pipeline
Pi 5 + AI HAT	Bio classifier NPU, OpenClaw	—
USC Laguna HPC	—	Fine-tuning, scoring, Monte Carlo (compute)
CMC HPC	—	vLLM hosting & deployment (hosting only)

When to use Gemini vs. Claude

Task	Model	Example message
Summarize a paper	Gemini	“Summarize arxiv.org/abs/2401.00000 ”
Morning project brief	Gemini	Automated 8am cron
Quick code question	Gemini	“What does <code>scale_pos_weight</code> do in XG-Boost?”
Find references	Gemini	“Find 3 papers on NLP and electoral forecasting”
Draft a paragraph	Gemini	“Draft an intro paragraph on portfolio theory”
Review pipeline logic	Claude	“@claude review kernels/shock.py”
Debug test failure	Claude	“@claude debug this pytest output: [paste]”
Write or fix LaTeX	Claude	“@claude fix the overflows in section 4”
Architecture decision	Claude	“@claude should I use Ridge or Lasso for the elasticity regression?”
Full session work	Claude	Open claude.ai directly

Useful commands

Command	Action
<code>openclaw doctor --fix</code>	Validate config and fix common issues
<code>openclaw skills install <name></code>	Install a skill from ClawHub
<code>openclaw logs</code>	View recent agent activity
<code>openclaw config edit</code>	Open config in editor
<code>@claude <message></code>	Route this message to Claude Sonnet
<code>@gemini <message></code>	Explicitly route to Gemini Pro

Cost guardrails — spend only where it matters

Service	Cost	Programme	Action
USC Laguna HPC	Free	USC student allocation	Use for training & compute
CMC HPC	Free	CMC student allocation	Use for hosting (vLLM) only
Modal GPU inference	~\$0.0003/call	\$30/mo free credit	~100k calls free
Vercel frontend	Free	Hobby tier	No action needed
GitHub	Free	Student Developer Pack	Apply at education.github.com
Namecheap domain	Free	via GitHub Student Pack	Claim via Student Pack
Gemini Pro	Free	Google AI Studio API	60 RPM, 1k req/day
Bluesky AT Protocol	Free	Open source	Primary social; full firehose
Apify X Scraper	Free	Free tier	Secondary live; 500 results/run
Meta Content Library	Free	Academic research	Apply Week 0
Claude Pro Max	Already paid	Existing subscription	Reserve for hard problems

Expected total cost during development: **\$0–\$5/month.**

Set a \$10/month hard cap on the Modal API at modal.com/settings/billing to prevent unexpected charges if demo traffic spikes. Do not use `keep_warm` on the Modal container during development — cold starts cost nothing and the \$30 free credit is sufficient for all testing. Reserve paid Modal time for the live demo day only.